Fundamental Knowledge on Quasi-Newton Methods

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Decide efficient allocations and plans, process signals and data with high accuracy, control robots and systems stably, purchase low-risk and high-return financial products, and learn model parameters from data. When we model the decision to be made as a decision variable x and a quantitative indicator representing its quality as an objective function $f(x)$, the best choice can be abstracted as an optimization problem. Mathematical optimization refers to such best choices, as well as the mathematical methods and academic fields for them. In this paper, we summarize the basic concepts of continuous optimization, particularly Newton's method and quasi-Newton methods, which are representative methods in this field.

1 Basic Concepts of Continuous Optimization

In this section, we explain the basic concepts of continuous optimization, including convexity, strong convexity, smoothness, and positive definiteness of the Hessian.

1.1 Convexity and Strong Convexity

The notions of convexity and strong convexity are fundamental in optimization theory. For simplicity, let us assume that f is a real-valued function defined over the entire space $\mathbb{R}^n$, i.e., $f\colon \mathbb{R}^n \to \mathbb{R}$. Also, let $\mu > 0$ be a constant. The function f is convex or μ-strongly convex if and only if the following hold for any $x, y \in \mathbb{R}^n$ and $\lambda \in [0,1]$:

$$\begin{aligned} &\text{(convex)} && f((1-\lambda)x+\lambda y) \le (1-\lambda)f(x)+\lambda f(y), \\ &(\mu\text{-strongly convex}) && f((1-\lambda)x+\lambda y) \le (1-\lambda)f(x)+\lambda f(y) - \frac{\mu}{2}\lambda(1-\lambda)\|x-y\|^2. \end{aligned}$$

There are more intuitive equivalent definitions of convexity and strong convexity using the gradient of f. Assume that f is of class C^1 and it takes real values over the domain $\mathbb{R}^n$. The function f is convex or μ-strongly convex if and only if the following hold for any $x, y \in \mathbb{R}^n$:

$$\begin{aligned} &\text{(convex)} && f(y) \ge f(x) + \nabla f(x)^\top (y-x), \\ &(\mu\text{-strongly convex}) && f(y) \ge f(x) + \nabla f(x)^\top (y-x) + \frac{\mu}{2}\|y-x\|^2. \end{aligned}$$

We show the equivalence of these two definitions below.

Proposition 1. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be of class C^1. Then the two definitions of convexity and μ-strong convexity are equivalent, respectively.*

Proof. We show the case of μ-strong convexity. First, assume that the former definition holds. By the definition of the gradient and the C^1 assumption on f, we have

$$\begin{aligned} \nabla f(x)^\top (y-x) &= \lim_{\lambda\to 0} \frac{f(x+\lambda(y-x)) - f(x)}{\lambda} \\ &\le \lim_{\lambda\to 0} \frac{1}{\lambda}\Big((1-\lambda)f(x)+\lambda f(y) - \frac{\mu}{2}\lambda(1-\lambda)\|x-y\|^2 - f(x)\Big) \\ &= f(y) - f(x) - \frac{\mu}{2}\|x-y\|^2. \end{aligned}$$

By rearranging the terms appropriately, we can show that the latter definition holds.

Conversely, assume that the latter definition holds. Let $z := (1-\lambda)x + \lambda y$. Then we have

$$f(x) \geq f(z) + \nabla f(z)^\top (x-z) + \frac{\mu}{2} \|x-z\|^2,$$
$$f(y) \geq f(z) + \nabla f(z)^\top (y-z) + \frac{\mu}{2} \|y-z\|^2.$$

Multiplying the first inequality by $1-\lambda$ and the second by λ, and adding them together, we deduce the following. Note that the terms involving $\nabla f(z)$ cancel out entirely by the definition of z.

$$\begin{aligned}
&(1-\lambda)f(x) + \lambda f(y) \\
&\geq f(z) + \nabla f(z)^\top ((1-\lambda)(x-z) + \lambda(y-z)) + \frac{\mu}{2}\left((1-\lambda)\|x-z\|^2 + \lambda\|y-z\|^2\right) \\
&= f(z) + \frac{\mu}{2}\left((1-\lambda)\lambda^2 \|x-y\|^2 + \lambda(1-\lambda)^2 \|y-x\|^2\right) \\
&= f(z) + \frac{\mu}{2}\lambda(1-\lambda)\|x-y\|^2.
\end{aligned}$$

Thus, the former definition holds. This shows that these definitions are equivalent.

By considering the case of $\mu = 0$, the convex case can be shown in the same way. □

Strong convexity indicates that, in addition to convexity, the objective function has uniformly positive curvature. We show examples of convex and strongly convex functions in Figure 1. Such definitions may vary slightly in the literature. For example, see [1, 2].

1.2 Positive Definiteness of the Hessian

Next, assuming that f is of class C^2, we show how convexity and strong convexity relate to the definiteness of the Hessian matrix $\nabla^2 f(x)$.

We define the definiteness of matrices beforehand. Let A be a symmetric matrix in $\mathbb{R}^{n\times n}$. A matrix A is called positive or negative definite (or semi-definite) based on the following conditions:

$$\begin{aligned}
&\text{(positive definite)} && v^\top A v > 0 && \forall v \in \mathbb{R}^n \setminus \{0\}, \\
&\text{(positive semi-definite)} && v^\top A v \geq 0 && \forall v \in \mathbb{R}^n, \\
&\text{(negative definite)} && v^\top A v < 0 && \forall v \in \mathbb{R}^n \setminus \{0\}, \\
&\text{(negative semi-definite)} && v^\top A v \leq 0 && \forall v \in \mathbb{R}^n.
\end{aligned}$$

A matrix is indefinite if it is neither positive nor negative definite. For matrices $A, B \in \mathbb{R}^{n\times n}$, the notation $A \succeq B$ indicates that $A - B$ is positive semi-definite. In particular, if B is a zero matrix, we may simply write $A \succeq 0$. We similarly define $\preceq$ for negative semi-definiteness. For $\mu \geq 0$, the condition $A \succeq \mu I$ is equivalent to $v^\top A v \geq \mu \|v\|^2$ for all $v \in \mathbb{R}^n$. It directly implies that all eigenvalues of A are at least μ. It also implies that the operator norm $\|A\|$ is at least μ, since the operator norm of a symmetric matrix is equal to an absolute value of its largest eigenvalue.

Here, strong convexity means that the function has uniformly positive curvature. This corresponds to the Hessian matrix also having uniformly positive curvature, i.e., being uniformly positive definite. In fact, it is known that convexity and strong convexity are equivalent to the definiteness of the Hessian matrix as follows.

Proposition 2. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be of class C^2. Then*

- *f is convex if and only if $\nabla^2 f(x) \succeq 0$ holds for all $x \in \mathbb{R}^n$.*
- *f is μ-strongly convex if and only if $\nabla^2 f(x) \succeq \mu I$ holds for all $x \in \mathbb{R}^n$.*

Proof. Firstly, assume that f is μ-strongly convex. For any $x \in \mathbb{R}^n$, $v \in \mathbb{R}^n$ and $t > 0$, letting $y = x \pm tv$ gives

$$\begin{cases} f(x+tv) \geq f(x) + t\nabla f(x)^\top v + \dfrac{\mu}{2} t^2 \|v\|^2, \\ f(x-tv) \geq f(x) - t\nabla f(x)^\top v + \dfrac{\mu}{2} t^2 \|v\|^2. \end{cases}$$

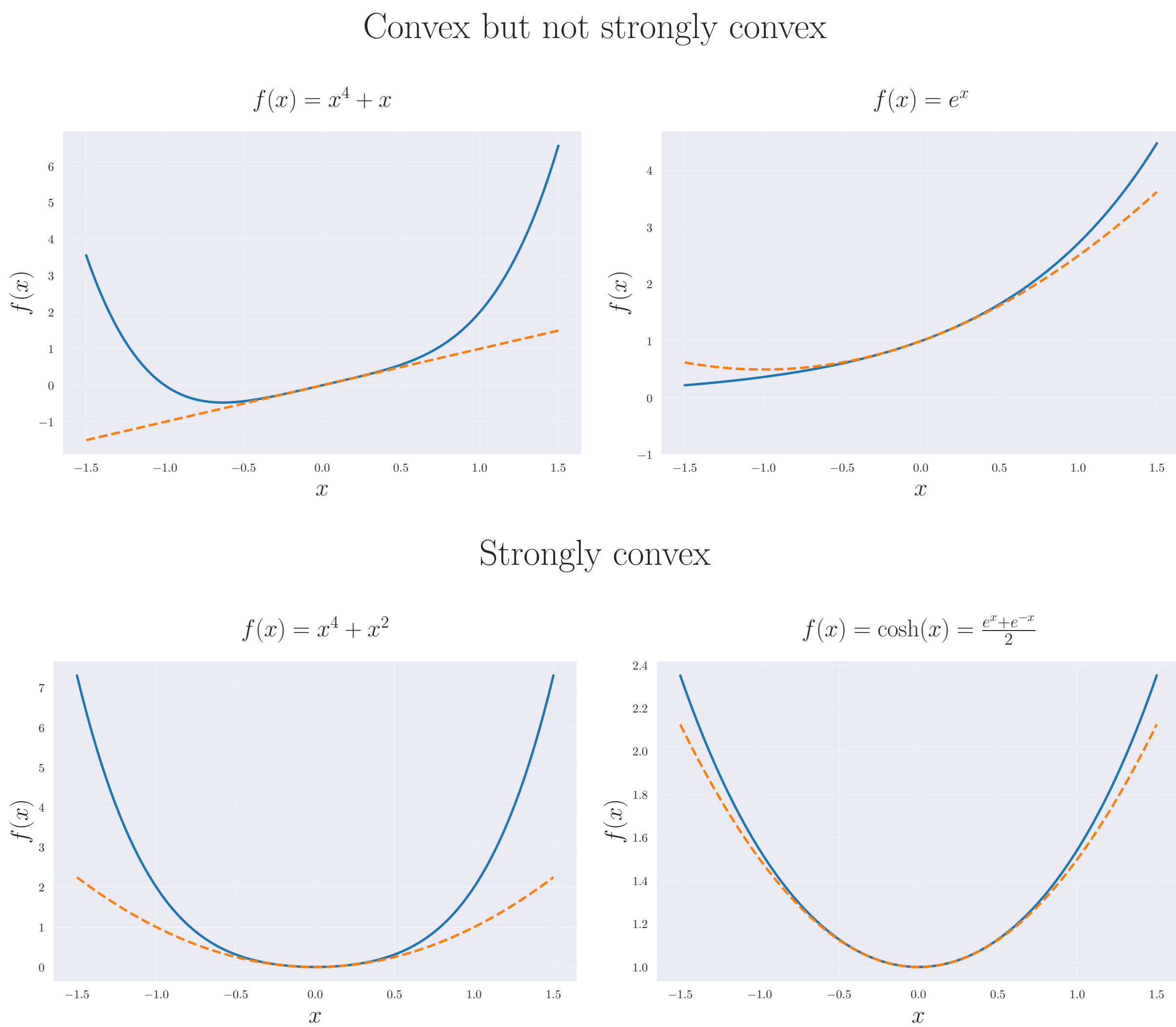

Figure 1: Examples of convex and strongly convex functions. The dashed line shows the quadratic approximation at $x = 0$. The upper two functions are convex but not strongly convex. The lower two functions are strongly convex, as there exists $\mu > 0$ satisfying the definition of strong convexity.

By Taylor's theorem, there exists $s_\pm \in (0,1)$ such that

$$\begin{cases} f(x+tv) = f(x) + t\nabla f(x)^\top v + \dfrac{1}{2}t^2 v^\top \nabla^2 f(x+s_+tv)v, \\ f(x-tv) = f(x) - t\nabla f(x)^\top v + \dfrac{1}{2}t^2 v^\top \nabla^2 f(x-s_-tv)v. \end{cases}$$

Combining these results gives

$$v^\top \frac{\nabla^2 f(x+s_+tv) + \nabla^2 f(x-s_-tv)}{2} v \geq \mu \left\| v \right\|^2.$$

Letting $t \to 0$ and using the continuity of $\nabla^2 f$ from the C^2 assumption gives

$$v^\top \nabla^2 f(x) v \geq \mu \left\| v \right\|^2.$$

Because $v \in \mathbb{R}^n$ is arbitrary, we obtain $\nabla^2 f(x) \succeq \mu I$.

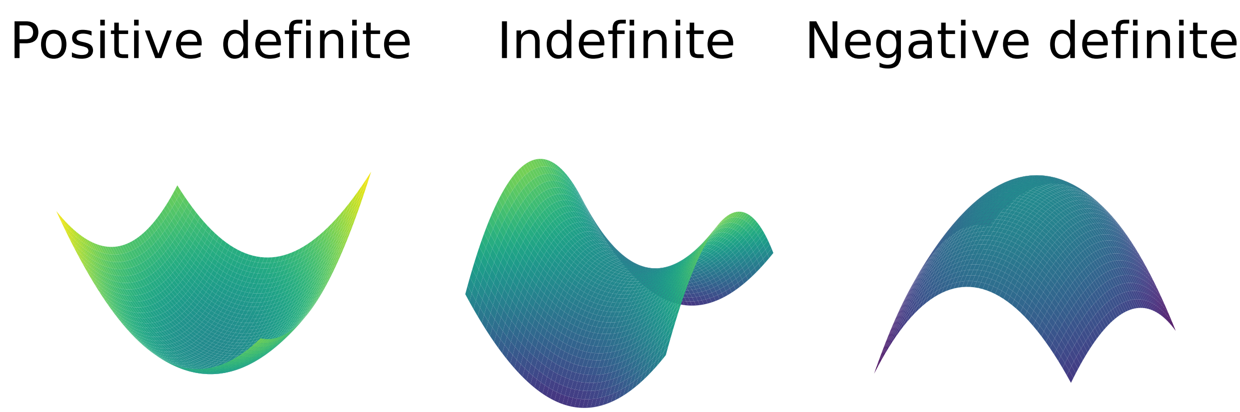

Figure 2: Quadratic function $f(x) = \frac{1}{2}(x-x_k)^\top H(x-x_k) + \nabla f(x_k)^\top(x-x_k) + f(x_k)$ in 2-dimensional space with Hessians H which is (left) positive definite, (center) indefinite, (right) negative definite. $f(x)$ is (left) convex, (center) nonconvex, and (right) concave, respectively.

Conversely, let $\mu > 0$ and assume $\nabla^2 f(x) \succeq \mu I$ for all $x \in \mathbb{R}^n$. By Taylor's theorem, for any $x, y \in \mathbb{R}^n$, letting $\phi(t) := f(x+t(y-x))$ gives

$$\begin{aligned} f(y) &= \phi(1) \\ &= \phi(0) + \phi'(0) + \int_0^1 (1-t)\phi''(t)\mathrm{d}t \\ &= f(x) + \nabla f(x)^\top(y-x) + \int_0^1 (1-t)(y-x)^\top \nabla^2 f(x+t(y-x))(y-x)\mathrm{d}t. \end{aligned}$$

By the assumption $\nabla^2 f(x) \succeq \mu I$, we have

$$\int_0^1 (1-t)(y-x)^\top \nabla^2 f(x+t(y-x))(y-x)\mathrm{d}t \geq \int_0^1 (1-t)\mu \|y-x\|^2 \mathrm{d}t = \frac{\mu}{2}\|y-x\|^2.$$

Thus, combining these results gives the definition of μ-strong convexity.

Setting $\mu = 0$ in the above shows the convex case in the same way. □

Quadratic functions whose Hessians are positive definite, indefinite, and negative definite are illustrated in Figure 2. We can confirm the correspondence between positive definiteness and convexity.

1.3 Smoothness

Finally, we introduce L-smoothness of functions. A function f is L-smooth if and only if there exists a constant $L > 0$ such that

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x-y\|$$

holds for all x, y. The L-smoothness of f is equivalent to the L-Lipschitz continuity of the gradient ∇f by definition. Please also refer to Figure 3.

In general, for nonconvex functions, the following upper bound holds.

Proposition 3. *Let $f: \mathbb{R}^n \to \mathbb{R}$ be an L-smooth function. Then for any $x, y \in \mathbb{R}^n$, we have*

$$f(y) \leq f(x) + \nabla f(x)^\top(y-x) + \frac{L}{2}\|y-x\|^2.$$

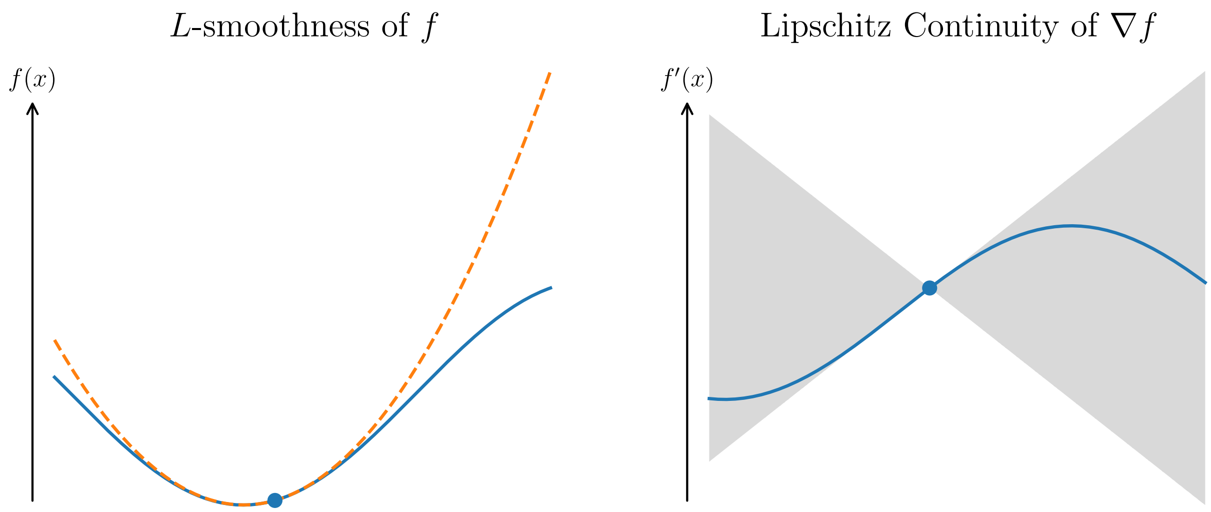

Figure 3: An example of L-smooth function in one-dimensional space. L-smoothness gives an upper bound of $f(x)$ as shown by the orange dashed line, and it is equivalent to the Lipschitz continuity of ∇f, which means that ∇f only exists in the gray area.

Proof. By the fundamental theorem of calculus, for any $x, y \in \mathbb{R}^n$, letting $\phi(t) := f(x + t(y-x))$ gives

$$\begin{aligned}
f(y) - f(x) = \phi(1) - \phi(0) &= \int_0^1 \phi'(t)\mathrm{d}t \\
&= \int_0^1 \nabla f(x + t(y-x))^\top (y-x)\mathrm{d}t \\
&= \nabla f(x)^\top (y-x) + \int_0^1 (\nabla f(x+t(y-x)) - \nabla f(x))^\top (y-x)\mathrm{d}t \\
&\leq \nabla f(x)^\top (y-x) + \int_0^1 \|\nabla f(x+t(y-x)) - \nabla f(x)\| \, \|y-x\| \,\mathrm{d}t && \text{(Cauchy–Schwarz inequality)} \\
&\leq \nabla f(x)^\top (y-x) + \int_0^1 Lt \, \|y-x\|^2 \,\mathrm{d}t && \text{(L-smoothness)} \\
&= \nabla f(x)^\top (y-x) + \frac{L}{2} \|y-x\|^2 .
\end{aligned}$$

This proves the upper bound. □

If f is convex in addition, the following lower bound also holds.

Proposition 4 ([3, Proposition 2.3.5])**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be a convex and L-smooth function. Then for any $x, y \in \mathbb{R}^n$, we have*

$$f(y) \geq f(x) + \nabla f(x)^\top (y-x) + \frac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2 .$$

Proof. Let us define the function $g_x(z)$ as follows.

$$\begin{aligned}
g_x(z) &:= f(z) - f(x) - \nabla f(x)^\top (z-x), \\
\nabla g_x(z) &= \nabla f(z) - \nabla f(x).
\end{aligned}$$

Since both $f(z)$ and the linear function $-f(x) - \nabla f(x)^\top (z-x)$ are convex in z, $g_x(z)$ is also convex and attains its minimum at $z = x$ where $\nabla g_x(z) = 0$ and thus the minimum value is $g_x(x) = 0$. Moreover, the L-smoothness of $f(z)$ directly implies the L-smoothness of $g_x(z)$. Thus, by the L-smoothness of $g_x(z)$, we have

$$0 \leq g_x\left(y - \frac{\nabla g_x(y)}{L}\right) \qquad \text{(minimum is $g_x(x) = 0$)}$$

$$
\begin{aligned}
&\le g_x(y) - (\nabla g_x(y))^\top \frac{\nabla g_x(y)}{L} + \frac{L}{2}\left\|\frac{\nabla g_x(y)}{L}\right\|^2 && \text{(by L-smoothness)}\\
&= g_x(y) - \frac{1}{2L}\|\nabla g_x(y)\|^2\\
&= f(y) - f(x) - \nabla f(x)^\top (y-x) - \frac{1}{2L}\|\nabla f(y) - \nabla f(x)\|^2. && \text{(definition of g_x)}
\end{aligned}
$$

This proves the lower bound. □

As we have seen so far, L-smoothness means that the rate of gradient change is bounded from above, and this can also be related to the definiteness of the Hessian matrix. In other words, as shown in the following proposition, the Hessian matrix is bounded by $-LI$ and $+LI$.

Proposition 5. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be of class C^2. Then f is L-smooth if and only if $-LI \preceq \nabla^2 f(x) \preceq +LI$ holds for all $x \in \mathbb{R}^n$.*

Proof. Since f is of class C^2, by the fundamental theorem of calculus, for any $x, y \in \mathbb{R}^n$, we have

$$
\nabla f(y) - \nabla f(x) = \int_0^1 \nabla^2 f(x + t(y-x))(y-x)\mathrm{d}t.
$$

Assume $-LI \preceq \nabla^2 f(x) \preceq +LI$ for all $x \in \mathbb{R}^n$. It implies that the operator norm of $\nabla^2 f(x)$ satisfies $\left\|\nabla^2 f(x)\right\| \le L$, and thus

$$
\begin{aligned}
\|\nabla f(y) - \nabla f(x)\| &= \left\|\int_0^1 \nabla^2 f(x+t(y-x))(y-x)\mathrm{d}t\right\| && \text{(previous equation)}\\
&\le \int_0^1 \left\|\nabla^2 f(x+t(y-x))(y-x)\right\| \mathrm{d}t && \text{(triangle inequality)}\\
&\le \int_0^1 L\|y-x\|\,\mathrm{d}t && \text{(by assumption)}\\
&= L\|y-x\|,
\end{aligned}
$$

which proves the L-smoothness of f.

Conversely, if f is L-smooth, then for any $x \in \mathbb{R}^n$ and $v \in \mathbb{R}^n$, we have

$$
\|\nabla f(x+tv) - \nabla f(x)\| \le L\|tv\| = Lt\|v\|.
$$

By Taylor's theorem, with remainder $r(t)$ satisfies $\|r(t)\|/t \to 0$ as $t \to 0$, we also have

$$
\nabla f(x+tv) - \nabla f(x) = t\nabla^2 f(x)v + r(t).
$$

Rearranging this gives

$$
\nabla^2 f(x)v = \lim_{t\to 0}\frac{\nabla f(x+tv) - \nabla f(x) - r(t)}{t} = \lim_{t\to 0}\frac{\nabla f(x+tv) - \nabla f(x)}{t}.
$$

Taking the inner product with v gives

$$
\begin{aligned}
v^\top \nabla^2 f(x) v &= \lim_{t\to 0}\left(\frac{\nabla f(x+tv) - \nabla f(x)}{t}\right)^\top v\\
&\le \lim_{t\to 0}\frac{\|\nabla f(x+tv) - \nabla f(x)\|}{t}\|v\| && \text{(Cauchy–Schwarz inequality)}\\
&\le \lim_{t\to 0}\frac{L\|tv\|}{t}\|v\| && \text{(by L-smoothness definition)}\\
&= L\|v\|^2.
\end{aligned}
$$

Because $v \in \mathbb{R}^n$ is arbitrary, we obtain $\nabla^2 f(x) \preceq LI$. In the same manner, $\nabla^2 f(x) \succeq -LI$ can also be shown, which completes the proof. □

1.3.1 Bound on Hessian Eigenvalues

By combining the discussions on L-smoothness and μ-strong convexity so far, we can also obtain bounds on the eigenvalues of the Hessian.

Proposition 6. *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be of class C^2. If f is L-smooth and μ-strongly convex, then the eigenvalues of the Hessian $\nabla^2 f(x)$ are contained in the interval $[\mu, L]$ for all $x \in \mathbb{R}^n$.*

Proof. By Propositions 2 and 5, for all $x \in \mathbb{R}^n$, we have

$$\mu I \preceq \nabla^2 f(x) \preceq LI.$$

This directly implies that the eigenvalues of $\nabla^2 f(x)$ are contained in the interval $[\mu, L]$. □

Thus, we can see that μ-strong convexity and L-smoothness correspond to the lower and upper bounds of the eigenvalues of the Hessian, respectively. The eigenvalues of the Hessian play a significant role in determining the convergence performance of algorithms, and such bounds can be utilized in various convergence analyses.

1.3.2 Baillon–Haddad Theorem

As an advanced topic, we introduce the following Baillon–Haddad theorem, which is one of the useful properties of L-smooth functions. Note that the C^2 assumption is not required in this theorem, and only the C^1 assumption is sufficient.

Proposition 7 (Baillon–Haddad theorem)**.** *Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be of class C^1. If f is L-smooth and convex, then for all $x, y \in \mathbb{R}^n$, ∇f is $1/L$-cocoercive, i.e.,*

$$(\nabla f(x) - \nabla f(y))^\top (x - y) \geq \frac{1}{L} \|\nabla f(x) - \nabla f(y)\|^2$$

for all $x, y \in \mathbb{R}^n$.

Proof. By the lower bound of f for convex and L-smooth functions shown above, we have

$$\begin{cases} f(y) \geq f(x) + \nabla f(x)^\top (y - x) + \dfrac{1}{2L} \|\nabla f(y) - \nabla f(x)\|^2, \\ f(x) \geq f(y) + \nabla f(y)^\top (x - y) + \dfrac{1}{2L} \|\nabla f(x) - \nabla f(y)\|^2. \end{cases}$$

Adding these two inequalities gives the Baillon–Haddad theorem. □

We also guide readers to other literature for the proof [4] [5, Proposition 12.60].

This theorem is sometimes used to derive the following inequality, which is useful for analyzing the update rules of quasi-Newton methods such as BFGS and L-BFGS methods. Assume that the function f is convex and L-smooth. For the sequences $\{x_k\}_k$ generated by an optimization algorithm, defining

$$s_k := x_{k+1} - x_k, \quad y_k := \nabla f(x_{k+1}) - \nabla f(x_k)$$

and applying Baillon–Haddad theorem with $x = x_{k+1}$ and $y = x_k$ yields

$$s_k^\top y_k \geq \frac{1}{L} \|y_k\|^2.$$

This inequality is sometimes used as one of the conditions for ensuring the positive definiteness of the approximate Hessian in quasi-Newton methods, and it plays an important role in theoretical guarantees.

2 Newton's Method

We define $f\colon \mathbb{R}^n \to \mathbb{R}$ as an objective function of class C^2, where n denotes the dimension of the decision variables. In this context, the following unconstrained optimization problem is one of the most fundamental optimization problems:

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x).$$

In this section, we outline Newton's method, which is a fundamental algorithm for this unconstrained optimization problem.

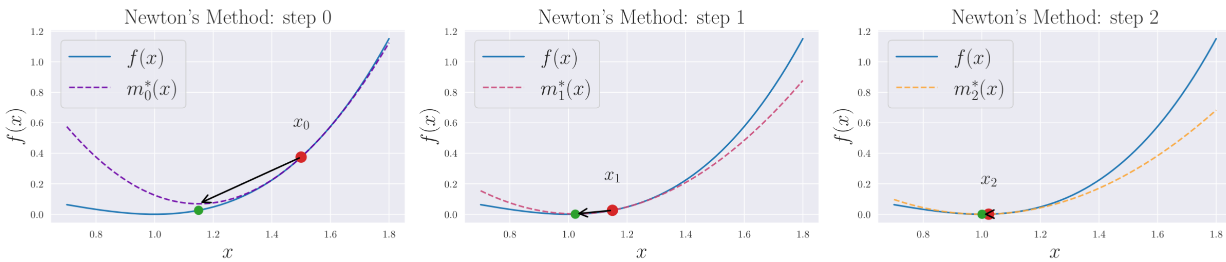

Figure 4: Illustration of iterations of Newton's method. In this example, $\alpha_k = 1$ holds for all steps, i.e., $x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} g_k$.

2.1 Algorithm of Newton's Method

Newton's method is a representative iterative algorithm that starts from an initial point x_0, successively updates the current point, and generates a sequence $\{x_k\}_k$. We assume that f is of class C^2 and strongly convex, so that the Hessian matrix $\nabla^2 f(x)$ is positive definite for all $x \in \mathbb{R}^n$, and thus invertible. Using the gradient $g_k := \nabla f(x_k)$ and Hessian $\nabla^2 f(x_k)$ at the k-th iterate x_k, the one step of Newton's method can be expressed as follows:

$$x_{k+1} \leftarrow x_k - \nabla^2 f(x_k)^{-1} g_k.$$

We can derive this update rule as follows. Firstly, the quadratic model of f at x_k is given by the Taylor expansion as follows:

$$m_k^*(x) := f(x_k) + g_k^\top (x - x_k) + \frac{1}{2}(x - x_k)^\top \nabla^2 f(x_k)(x - x_k).$$

The gradient of this model is

$$\nabla m_k^*(x) = g_k + \nabla^2 f(x_k)(x - x_k).$$

Since the quadratic model is strongly convex by assumption, a point $x^* \in \mathbb{R}^n$ is the minimizer of m_k^* if and only if $\nabla m_k^*(x) = 0$. Thus, solving this equation yields

$$\arg\min_{x \in \mathbb{R}^n} m_k^*(x) = x_k - \left(\nabla^2 f(x_k)\right)^{-1} g_k.$$

Thus, it is natural to select this x^* as the next iterate, and this leads to the update rule of Newton's method.

However, as we will see later, it does not generally guarantee global convergence. Thus, we often introduce a step size $\alpha_k > 0$ determined by line search to ensure a sufficient decrease in function values, leading to the update rule as follows:

$$x_{k+1} \leftarrow x_k - \alpha_k \nabla^2 f(x_k)^{-1} g_k.$$

This variant is called the damped Newton's method or relaxed Newton's method, and it is a common practice to use this version in general optimization problems.

The update direction $d_k := -\nabla^2 f(x_k)^{-1} g_k$ is known as the Newton direction. When the Hessian is positive definite, the following relation shows that it is a descent direction, meaning that it is a direction in which the objective function decreases.

$$g_k^\top d_k = -g_k^\top \nabla^2 f(x_k)^{-1} g_k < 0.$$

When f is nonconvex, meaning that the Hessian $\nabla^2 f(x_k)$ is negative definite or indefinite, d_k may not be a descent direction, and Newton's method may fail to converge. The modified Newton's method [6, Sec. 3.4] is one remedy for handling such cases, but we omit the explanation here.

2.2 Convergence Properties of Newton's Method

We next present standard convergence properties for Newton's method.

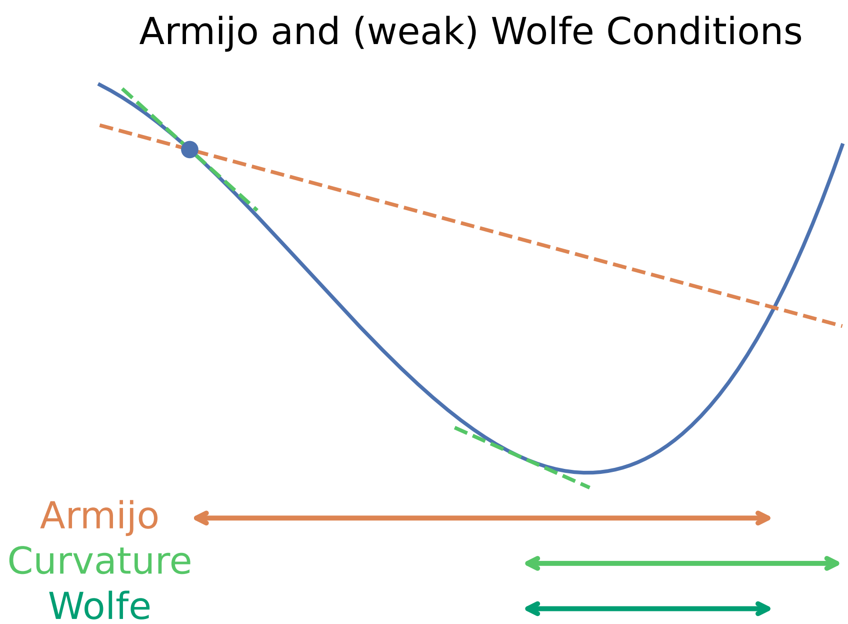

Figure 5: Illustration of the line search conditions. Each arrow indicates the range where the corresponding condition is satisfied.

2.2.1 Global Convergence

We first state a general global convergence result

$$\lim_{k\to\infty} \|g_k\| = 0$$

for methods with line search, and then explain its application to damped Newton's method.

We consider a general iterative method of the form

$$x_{k+1} \leftarrow x_k + \alpha_k d_k.$$

The direction d_k is a descent direction satisfying $g_k^\top d_k < 0$ and $\alpha_k > 0$ is the step size determined by line search. We employ the (weak) Wolfe conditions [6, Sec. 3.1] to determine the step size α_k as follows:

$$\begin{cases} f(x_k + \alpha_k d_k) & \le f(x_k) + c_1 \alpha_k g_k^\top d_k, \\ -g_{k+1}^\top d_k & \le -c_2 g_k^\top d_k. \end{cases}$$

$0 < c_1 < c_2 < 1$ are constants.

The first condition is known as the Armijo condition, which requires that the function value at the next point $f(x_k + \alpha_k d_k)$ decreases by at least a fraction c_1 of the predicted decrease $\alpha_k g_k^\top d_k$.

The second condition is known as the curvature condition. If d_k is a descent direction at the next point x_{k+1}, then $-g_{k+1}^\top d_k$ is positive, and the objective function value decreases if we move slightly in the direction of d_k. The curvature condition requires that this decrease rate is not too large, meaning that a sufficiently large step size is taken along d_k. In the weak Wolfe conditions, it is also possible that the step size is too long, but in the strong Wolfe conditions, by taking the absolute value, it requires that an appropriate step size is taken.

The Wolfe conditions refer to the simultaneous satisfaction of Armijo and curvature conditions. Figure 5 illustrates the ranges where these conditions are satisfied.

We also define $\theta_k \in [0, \pi]$ as the angle between the direction d_k and the negative gradient $-g_k$, satisfying

$$\cos\theta_k = \frac{-g_k^\top d_k}{\|g_k\| \, \|d_k\|}.$$

Under these settings, we present a simplified version of the following classical well-known result.

Theorem 1 ([6, Theorem 3.2]). *Suppose that $f\colon \mathbb{R}^n \to \mathbb{R}$ is of class C^1 and bounded below in $\mathbb{R}^n$, and L-smooth. Consider the following iterative method*

$$x_{k+1} \leftarrow x_k + \alpha_k d_k.$$

It starts from an initial point $x_0 \in \mathbb{R}^n$, and the step size α_k satisfies the Wolfe conditions. For the angles $\{\theta_k\}_k$, if there exists a positive constant δ such that $\cos\theta_k \geq \delta > 0$ for all k, then the method generates a sequence $\{x_k\}_k$ satisfying

$$\lim_{k\to\infty} \|g_k\| = 0.$$

Proof. From the Wolfe conditions, Cauchy–Schwarz inequality and the L-smoothness of f, the following two relations hold:

$$\begin{cases} (g_{k+1}-g_k)^\top d_k = g_{k+1}^\top d_k - g_k^\top d_k \geq (c_2-1) g_k^\top d_k, \\ (g_{k+1}-g_k)^\top d_k \leq \|g_{k+1}-g_k\| \|d_k\| \leq L \|x_{k+1}-x_k\| \|d_k\| = \alpha_k L \|d_k\|^2. \end{cases}$$

By combining these two relations, we obtain

$$\alpha_k \geq \frac{c_2-1}{L} \frac{g_k^\top d_k}{\|d_k\|^2}.$$

Thus, since $\cos\theta_k > 0$ implies that d_k is a descent direction, i.e., $g_k^\top d_k < 0$, we have the following inequality.

$$\begin{aligned} f(x_{k+1}) &\leq f(x_k) + c_1 \alpha_k g_k^\top d_k && \text{(Armijo condition)} \\ &\leq f(x_k) - c_1 \frac{1-c_2}{L} \frac{(g_k^\top d_k)^2}{\|d_k\|^2} && \text{(previous inequality)} \\ &= f(x_k) - c_1 \frac{1-c_2}{L} \cos^2\theta_k \|g_k\|^2. && \text{(definition of } \theta_k\text{)} \end{aligned}$$

By repeatedly applying this inequality, we obtain

$$f(x_{k+1}) \leq f(x_0) - c_1 \frac{1-c_2}{L} \sum_{j=0}^{k} \cos^2\theta_j \|g_j\|^2.$$

Since we assumed that f is bounded below, there exists a constant f^* such that $f(x_{k+1}) \geq f^*$ for all k. Hence, we obtain the following condition, known as the Zoutendijk condition:

$$\sum_{k=0}^{\infty} \cos^2\theta_k \|g_k\|^2 \leq \frac{L}{c_1(1-c_2)} (f(x_0) - f^*) < \infty.$$

This condition implies that

$$\cos^2\theta_k \|g_k\|^2 \to 0.$$

Combined with the assumption that $\cos\theta_k \geq \delta > 0$ for all k, it follows immediately that

$$\lim_{k\to\infty} \|g_k\| = 0.$$

This completes the proof. □

Since damped Newton's method is a special case of the setting in Theorem 1 with $d_k = -\nabla^2 f(x_k)^{-1} g_k$, we can apply this result under appropriate conditions of f. In particular, if f is L-smooth and μ-strongly convex, then for any k, we have

$$\cos\theta_k = \frac{g_k^\top \nabla^2 f(x_k)^{-1} g_k}{\|g_k\| \|\nabla^2 f(x_k)^{-1} g_k\|} \geq \frac{\|g_k\|^2 \lambda_{\min}(\nabla^2 f(x_k)^{-1})}{\|g_k\|^2 \lambda_{\max}(\nabla^2 f(x_k)^{-1})} \geq \frac{\mu}{L}.$$

Here, $\lambda_{\min}(\cdot)$ and $\lambda_{\max}(\cdot)$ denote the minimum and maximum eigenvalues, respectively. Thus, by setting $\delta = \mu/L$ in Theorem 1, we obtain the global convergence of damped Newton's method for strongly convex and smooth functions under line search satisfying the Wolfe conditions.

2.2.2 Local Quadratic Convergence

We next present a classical result on the local convergence of Newton's method. We again provide a simplified version for brevity.

Theorem 2 ([6, Theorem 3.5])**.** *Suppose that the Hessian matrix $\nabla^2 f(x)$ is Lipschitz continuous in a neighborhood of the solution x^*, and that sufficient second-order optimality conditions hold (i.e., $\nabla f(x^*) = 0$ and $\nabla^2 f(x^*)$ is positive definite). In Newton's method, if $\alpha_k = 1$ holds for all k and the initial point x_0 is sufficiently close to x^*, then the sequence of gradient norms $\{\|\nabla f(x_k)\|\}_k$ converges quadratically.*

Proof. Let $k = 0$ and consider the update from x_k to x_{k+1}. Since the Hessian $\nabla^2 f(x_k)$ is invertible in a sufficiently small neighborhood of x^*, from the optimality condition $\nabla f(x^*) = 0$ and the assumption $\alpha_k = 1$, we have

$$\begin{aligned} x_{k+1} - x^* &= x_k - x^* - \left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k) \\ &= \left(\nabla^2 f(x_k)\right)^{-1} \left(\nabla^2 f(x_k)(x_k - x^*) - (\nabla f(x_k) - \nabla f(x^*))\right). \end{aligned}$$

By Taylor's theorem and the triangle inequality, we have

$$\begin{aligned} &\left\|\nabla^2 f(x_k)(x_k - x^*) - (\nabla f(x_k) - \nabla f(x^*))\right\| \\ &= \left\|\nabla^2 f(x_k)(x_k - x^*) - \int_0^1 \nabla^2 f(x_k + t(x^* - x_k))(x_k - x^*)\mathrm{d}t\right\| \\ &\le \int_0^1 \left\|\nabla^2 f(x_k) - \nabla^2 f(x_k + t(x^* - x_k))\right\| \|x_k - x^*\| \,\mathrm{d}t. \end{aligned}$$

If $\nabla^2 f$ is Lipschitz continuous with constant L^{H}, then the integrand is bounded by $L^{\mathrm{H}} t \|x_k - x^*\|$, and integrating yields

$$\left\|\nabla^2 f(x_k)(x_k - x^*) - (\nabla f(x_k) - \nabla f(x^*))\right\| \le \frac{1}{2} L^{\mathrm{H}} \|x_k - x^*\|^2.$$

Since $\nabla^2 f(x^*)$ is positive definite, there exists a radius $r > 0$ such that for all x satisfying $\|x - x^*\| \le r$,

$$\left\|\left(\nabla^2 f(x)\right)^{-1}\right\| \le 2 \left\|\left(\nabla^2 f(x^*)\right)^{-1}\right\|$$

holds. Combining these results, we obtain

$$\begin{aligned} \|x_{k+1} - x^*\| &\le \left\|\left(\nabla^2 f(x_k)\right)^{-1}\right\| \left\|\nabla^2 f(x_k)(x_k - x^*) - (\nabla f(x_k) - \nabla f(x^*))\right\| \\ &\le L^{\mathrm{H}} \left\|\left(\nabla^2 f(x^*)\right)^{-1}\right\| \|x_k - x^*\|^2. \end{aligned}$$

Let $\widetilde{L} := L^{\mathrm{H}} \left\|\left(\nabla^2 f(x^*)\right)^{-1}\right\|$. If the initial point is chosen such that $\|x_0 - x^*\| \le \min\{r, 1/(2\widetilde{L})\}$, then by induction $\{x_k\}_k$ remains within the neighborhood and converges to x^*. Thus, the above error bound implies quadratic convergence of $\{x_k\}_k$.

To show quadratic convergence of the gradient norm, we use $x_{k+1} - x_k = -\left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k)$ and $\nabla f(x_k) + \nabla^2 f(x_k)(x_{k+1} - x_k) = 0$, yielding

$$\begin{aligned} \|\nabla f(x_{k+1})\| &= \left\|\nabla f(x_{k+1}) - \nabla f(x_k) - \nabla^2 f(x_k)(x_{k+1} - x_k)\right\| \\ &\le \int_0^1 \left\|\nabla^2 f(x_k + t(x_{k+1} - x_k)) - \nabla^2 f(x_k)\right\| \|x_{k+1} - x_k\| \,\mathrm{d}t \\ &\le \frac{1}{2} L^{\mathrm{H}} \|x_{k+1} - x_k\|^2 \\ &\le \frac{1}{2} L^{\mathrm{H}} \left\|\left(\nabla^2 f(x_k)\right)^{-1}\right\|^2 \|\nabla f(x_k)\|^2 \\ &\le 2L^{\mathrm{H}} \left\|\left(\nabla^2 f(x^*)\right)^{-1}\right\|^2 \|\nabla f(x_k)\|^2. \end{aligned}$$

Therefore, $\|\nabla f(x_k)\|$ converges quadratically to 0. □

These Theorems 1 and 2 indicate that Newton's method has both global convergence and local quadratic convergence properties under appropriate conditions. This rapid convergence is a significant advantage of Newton's method compared to first-order methods such as gradient descent, which typically exhibit only linear convergence.

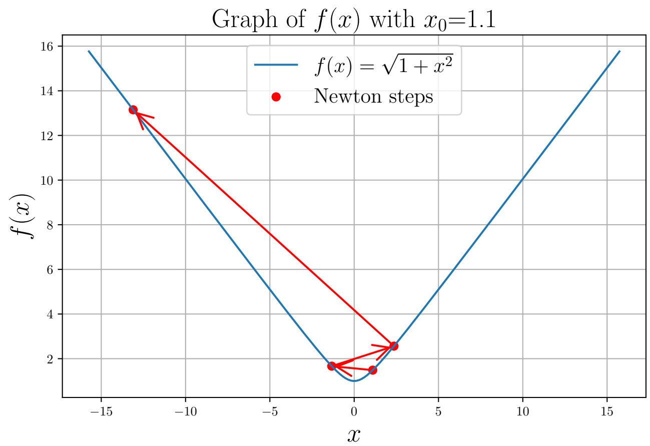

Figure 6: An example where Newton's method diverges with initial point $x_0 = 1.1$

2.3 Elements for Global Convergence

As mentioned earlier, the positive definiteness of the Hessian matrix and the selection of the step size via line search play important roles in Newton's method. In this subsection, we explain why these elements are required.

2.3.1 Positive Definiteness of the Hessian Matrix

We have assumed that f is strongly convex so far, meaning that the Hessian matrix $\nabla^2 f(x_k)$ is positive definite at each iteration. This assumption is crucial for Newton's method to converge locally to an optimal solution. When the Hessian matrix is positive definite, Newton's method provides a descent direction, since

$$\nabla f(x_k)^\top d_k = -\nabla f(x_k)^\top \left(\nabla^2 f(x_k)\right)^{-1} \nabla f(x_k) < 0.$$

In contrast, when the Hessian matrix is negative definite or indefinite, a decrease in the function value is not guaranteed, and Newton's method can move away from the optimal solution. In the indefinite case, the function value may decrease, but the risk of converging to a saddle point is also increased. Therefore, the positive definiteness of the Hessian matrix is important when applying Newton's method.

2.3.2 Issues with Full Steps

Newton's method with $\alpha_k = 1$ at every iteration may lead to problematic behavior. Here, we present specific examples of such failures and discuss the necessity of line search as a countermeasure.

An Example Where Newton's Method Diverges Consider the following function:

$$\begin{aligned} f(x) &:= \sqrt{1+x^2}, \\ f'(x) &= \frac{x}{\sqrt{1+x^2}}, \\ f''(x) &= \frac{1}{(1+x^2)^{3/2}}. \end{aligned}$$

When the absolute value of the initial point exceeds 1, Newton's method diverges for this function, as shown in Figure 6.

This is because, at points far from the optimal solution $x^* = 0$, the value of the second derivative, i.e., the eigenvalue of the Hessian matrix, becomes very small, leading to an excessively large Newton step.

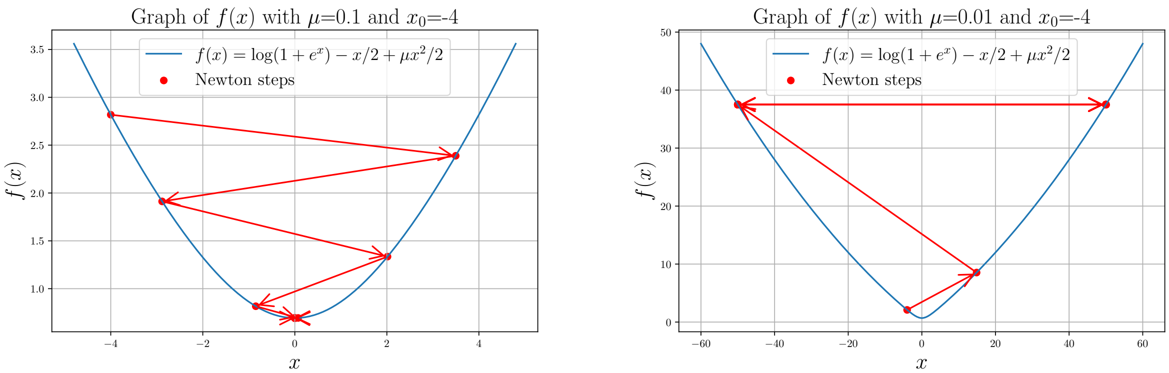

Figure 7: (Left) Newton's method converges for $x_0 = -4,\ \mu = 0.1$. (Right) Newton's method oscillates for $x_0 = -4,\ \mu = 0.01$.

Oscillation of Newton's Method for Strongly Convex Functions In the previous example, the objective function was not strongly convex and did not necessarily possess favorable properties. However, even for functions with the strong convexity property, there exist examples where Newton's method fails to converge [7, Example 1.4.3].

In this example, for $\mu > 0$, consider the function

$$
\begin{aligned}
f(x) &:= \log(1+e^x) - \frac{x}{2} + \frac{\mu x^2}{2}, \\
f'(x) &= \frac{e^x}{1+e^x} - \frac{1}{2} + \mu x, \\
f''(x) &= \frac{e^x}{(1+e^x)^2} + \mu, \\
f'''(x) &= \frac{e^x(1-e^x)}{(1+e^x)^3}, \\
f''''(x) &= \frac{e^x(1-4e^x+e^{2x})}{(1+e^x)^4}.
\end{aligned}
$$

This function has the following properties:

- It is μ-strongly convex.
- $\max_x |f''(x)| = \frac{1}{4} + \mu$ (attained at $e^x = 1$). Hence, ∇f is L-smooth with $L = \frac{1}{4} + \mu$.
- $\max_x |f'''(x)| = \frac{1}{6\sqrt{3}}$ (attained at $e^x = 2 - \sqrt{3}$). Hence, $\nabla^2 f$ is L^{H}-Lipschitz with $L^{\mathrm{H}} = \frac{1}{6\sqrt{3}}$.

Nevertheless, when the initial point x_0 is sufficiently large relative to μ, Newton's method exhibits oscillatory behavior, as shown in Figure 7.

Necessity of Line Search From these examples, it is clear that adopting a full step size $\alpha_k = 1$ in Newton's method can lead to divergence or oscillation, even for strongly convex and smooth functions. Therefore, it is essential to select the step size α_k appropriately to ensure convergence. This is why damped Newton's method with line search is often used.

2.4 Comparison with Newton's Method as a Root-Finding Algorithm

As a conceptual supplement, we briefly clarify the relationship between "Newton's method in optimization" and "Newton's method as a root-finding algorithm." These two formulations are closely related, but they have different perspectives and applications.

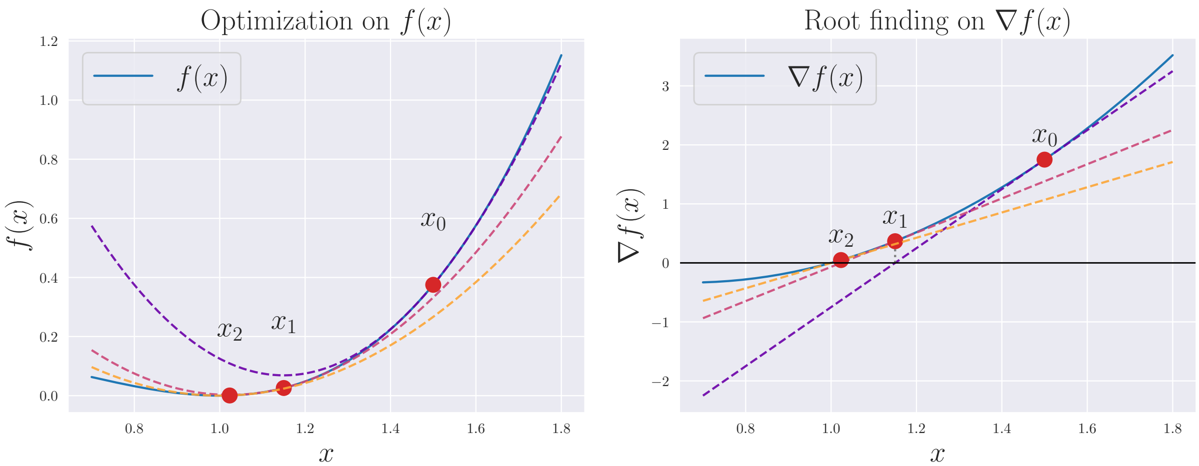

Figure 8: Optimization of the function $f(x) = x^3 - 2x^2 + x$ and root-finding for the gradient $\nabla f(x) = 3x^2 - 4x + 1$.

2.4.1 Newton's Method in Optimization

In the context of optimization, Newton's method typically refers to an algorithm for finding a local minimizer of a twice-differentiable function $f\colon \mathbb{R}^n \to \mathbb{R}$. The update of Newton's method with a step size $\alpha_k = 1$ was given by

$$x_{k+1} = x_k - \nabla^2 f(x_k)^{-1} \nabla f(x_k).$$

In the scalar case, the above expression reduces to $x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$.

2.4.2 Newton's Method as a Root-Finding Algorithm

When referring to Newton's method (or the Newton–Raphson method), one may mean a root-finding algorithm for a differentiable scalar function $g\colon \mathbb{R} \to \mathbb{R}$ that solves $g(x) = 0$. It is also arguably fundamental and widely known.

Starting from an initial value x_0, the iteration is given by

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)}.$$

Note that in the scalar case, since $g(x_k) = f'(x_k)$, this is the same formula as the update of Newton's method in optimization we saw earlier. Geometrically, this corresponds to approximating the graph of g near x_k by its tangent line and selecting the intersection of this tangent with the x-axis as the next approximate solution.

2.4.3 Relationship Between the Two Formulations

Let us examine the equivalence of the two formulations through a concrete example. Figure 8 illustrates the relationship between these formulations using a simple cubic function $f(x)$. The left figure corresponds to optimization of $f(x)$, while the right figure corresponds to root-finding for $g(x) = \nabla f(x)$. In the optimization formulation, $f(x) = x^3 - 2x^2 + x$ is approximated by its second-order Taylor expansion

$$m_k^*(x) = f(x_k) + \nabla f(x_k)(x - x_k) + \frac{1}{2} \nabla^2 f(x_k)(x - x_k)^2$$

and the minimizer of this quadratic model is taken as x_{k+1}. In the root-finding formulation, $\nabla f(x) = 3x^2 - 4x + 1$ is approximated by its tangent line, namely

$$\nabla m_k^*(x) = \nabla f(x_k) + \nabla^2 f(x_k)(x - x_k)$$

and the root of this linear model is chosen as x_{k+1}. Thus, under the assumption of positive definiteness, it is clear that both formulations perform essentially equivalent operations, and their correspondence can be explicitly verified.

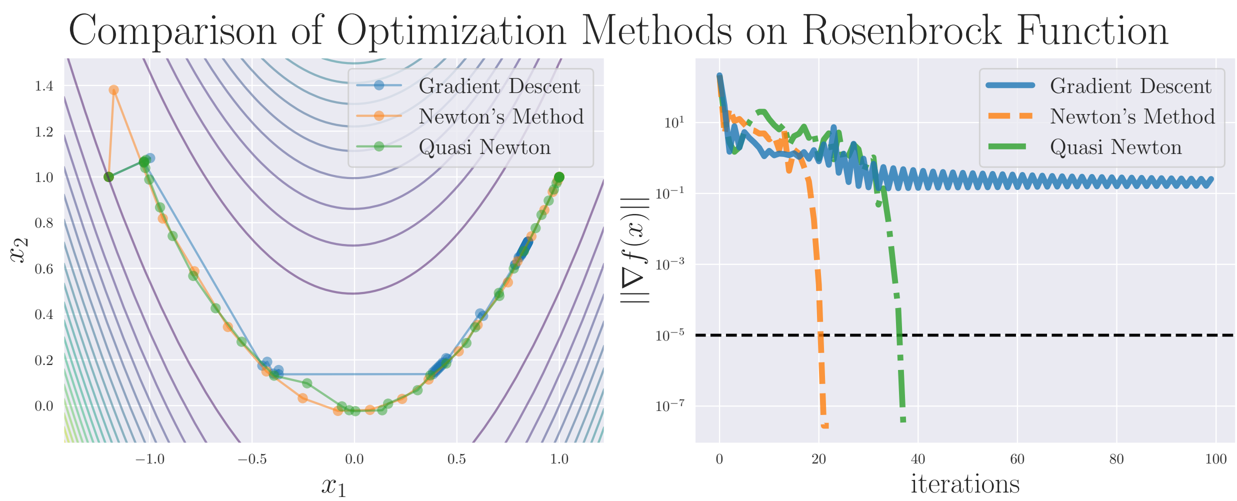

Figure 9: Comparison of Newton's method, quasi-Newton methods, and gradient descent.

3 Quasi-Newton Methods

In this section, we discuss quasi-Newton methods. Quasi-Newton methods are based on Newton's method but aim to reduce its primary drawback, namely the high computational cost of evaluating the Hessian matrix. Instead of using the true Hessian matrix $\nabla^2 f(x_k)$, an approximation matrix B_k is employed, thereby achieving fast convergence while keeping the computational cost low.

Figure 9 shows a comparison of Newton's method, quasi-Newton methods, and gradient descent. Gradient descent updates the iterate in the direction opposite to the gradient. Although its per-iteration computational cost is the lowest, its convergence is slow. Newton's method requires the fewest iterations, but it has the highest computational cost per step. Quasi-Newton methods lie between these two extremes and strike a balance between them. In particular, when the methods are compared in terms of total computation time rather than the number of iterations, quasi-Newton methods often exhibit the best performance.

Line-search-based quasi-Newton methods generate a sequence $\{x_k\}_k$ that converges to the optimal solution x^* as follows:

$$x_{k+1} = x_k - \alpha_k B_k^{-1} \nabla f(x_k) = x_k - \alpha_k H_k \nabla f(x_k).$$

Here, $\alpha_k > 0$ is the step size determined by line search, B_k is an approximation of the Hessian matrix $\nabla^2 f(x_k)$ at the point x_k, and $H_k := B_k^{-1}$ denotes its inverse. This update rule closely resembles that of Newton's method, which is why it is called a quasi-Newton method.

The core of quasi-Newton methods lies in how to update B_k at each iteration so that it approaches the true Hessian matrix. Figure 10 illustrates this concept. First, around the current point x_k, a quadratic approximation model of the objective function f is constructed using B_k. Next, this quadratic model is minimized to obtain the next point x_{k+1}. After obtaining x_{k+1}, the approximation matrix B_k is updated to B_{k+1} using the gradient information at x_k and x_{k+1}. This procedure is repeated until convergence, which constitutes the quasi-Newton method.

3.1 Secant Condition

Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be of class C^2. Suppose that a symmetric matrix B_k is given as an approximation of the Hessian matrix $\nabla^2 f(x_k)$. We consider updating the approximate Hessian to B_{k+1} for the next point x_{k+1}. Although there are infinitely many possible candidates for such a matrix, it is natural to impose symmetry on B_{k+1}, since the true Hessian matrix is symmetric. For the current point x_k and the next point x_{k+1}, we define the step and the gradient difference as follows:

$$s_k := x_{k+1} - x_k, \qquad y_k := \nabla f(x_{k+1}) - \nabla f(x_k).$$

In the following, we generally assume that $s_k \neq 0$ and $y_k^\top s_k \neq 0$. Note that the s in s_k stands for "step." Since the new approximation is constructed around x_{k+1}, we approximate $\nabla f(x_k)$ using the Hessian matrix $\nabla^2 f(x_{k+1})$ and its

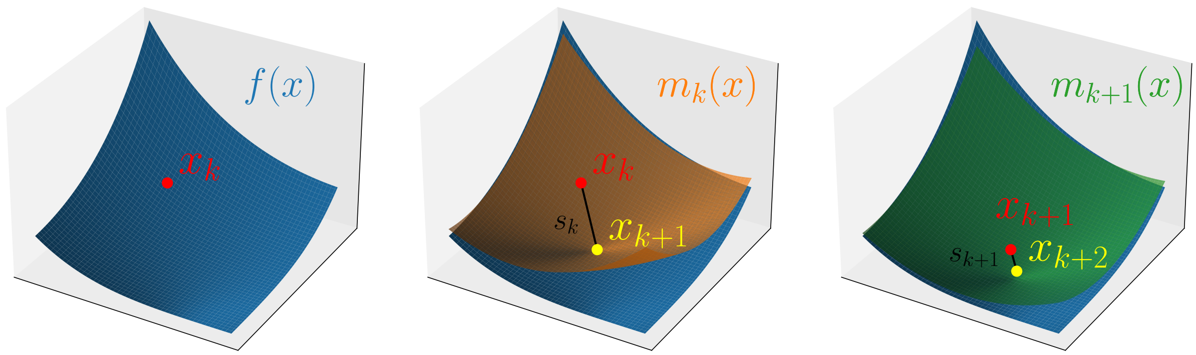

Figure 10: Conceptual illustration of quasi-Newton methods. (1) Given the objective function f (blue surface) and the current point x_k (red point). (2) Move to the minimizer x_{k+1} (yellow cross) of the quadratic model $m_k(x)$ (orange surface) constructed using the current approximate Hessian. (3) Update the quadratic model (green surface) and repeat this procedure.

approximation B_{k+1} as follows:

$$\begin{aligned}\nabla f(x_k) &\approx \nabla f(x_{k+1}) + \nabla^2 f(x_{k+1})(x_k - x_{k+1}) \\ &\approx \nabla f(x_{k+1}) + B_{k+1}(x_k - x_{k+1}).\end{aligned}$$

Requiring the above approximation to hold as an equality yields

$$B_{k+1}(x_{k+1} - x_k) = \nabla f(x_{k+1}) - \nabla f(x_k).$$

Equivalently, this can be rewritten as

$$B_{k+1}s_k = y_k.$$

This relation is called the secant condition or the quasi-Newton equation.

More precisely, the secant condition can also be justified as follows. Consider a quadratic approximation model around x_{k+1}:

$$m_{k+1}(x) = f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}(x - x_{k+1}).$$

By construction, regardless of B_{k+1}, this model satisfies

$$\begin{cases} m_{k+1}(x_{k+1}) = f(x_{k+1}), \\ \nabla m_{k+1}(x_{k+1}) = \nabla f(x_{k+1}). \end{cases}$$

In addition, the secant condition ensures that the model satisfies

$$\begin{aligned}\nabla m_{k+1}(x_k) &= \nabla f(x_{k+1}) - B_{k+1}(x_{k+1} - x_k) && \text{(definition of the model)} \\ &= \nabla f(x_{k+1}) - (\nabla f(x_{k+1}) - \nabla f(x_k)) && \text{(secant condition)} \\ &= \nabla f(x_k).\end{aligned}$$

This implies that the secant condition ensures that the quadratic model $m_{k+1}(x)$ accurately reflects the known information at both x_{k+1} and x_k, except for the previous function value $f(x_k)$.

3.2 Representative Quasi-Newton Update Rules

Given B_k, s_k, and y_k, there exist many update rules that produce B_{k+1} satisfying the secant condition. We introduce several representative ones together with their derivations [8]. In this subsection only, for brevity, we abbreviate B_k, B_{k+1}, s_k, and y_k as B, $\bar{B}$, s, and y, respectively.

3.2.1 Broyden's Update

Broyden's update is one of the oldest known update rules, but it is not often used for quasi-Newton methods in practice due to its lack of symmetry. We introduce it briefly for historical context. The update formulas are given by

$$\bar{B}_{\text{Broyden}} := B + \frac{(y - Bs)s^\top}{s^\top s}.$$

Derivation (Broyden) We derive this formula from simple structural assumptions [8, Section 4].

Proposition 8. *Assume that $\bar{B}$ satisfies the secant condition*

$$\bar{B}s = y,$$

and the following constraint for all vectors $z \in \mathbb{R}^n$ such that $z^\top s = 0$:

$$\bar{B}z = Bz.$$

Then $\bar{B}$ is uniquely determined, and it is $\bar{B}_{\text{Broyden}}$.

Proof. A vector s and a basis for the orthogonal complement of s form a basis of $\mathbb{R}^n$. Since the conditions on $\bar{B}$ completely determine the action of $\bar{B}$ with respect to this basis, $\bar{B}$ is uniquely determined.

We now show that $\bar{B}_{\text{Broyden}}$ satisfies the imposed conditions. Let z be any vector satisfying $z^\top s = 0$. Then

$$\begin{aligned}
\bar{B}_{\text{Broyden}} s &= \left(B + \frac{(y - Bs)s^\top}{s^\top s}\right) s = Bs + (y - Bs) = y, \\
\bar{B}_{\text{Broyden}} z &= \left(B + \frac{(y - Bs)s^\top}{s^\top s}\right) z = Bz + (z^\top s)\frac{y - Bs}{s^\top s} = Bz.
\end{aligned}$$

Hence $\bar{B}_{\text{Broyden}}$ satisfies the imposed conditions.

Therefore, by uniqueness, we conclude that $\bar{B} = \bar{B}_{\text{Broyden}}$. □

Broyden's update can also be characterized as a minimal-change update in the Frobenius norm.

Proposition 9 ([8, Theorem 4.1])**.** *Let $B \in \mathbb{R}^{n \times n}$, $y \in \mathbb{R}^n$, and $s \in \mathbb{R}^n \setminus \{0\}$ be given. Then $\bar{B}_{\text{Broyden}}$ is the unique solution of the following optimization problem:*

$$\begin{aligned}
\underset{\tilde{B} \in \mathbb{R}^{n \times n}}{\text{minimize}} \quad & \left\|\tilde{B} - B\right\|_F \\
\text{subjectto} \quad & \tilde{B}s = y.
\end{aligned}$$

Proof. Instead of directly solving the optimization problem, we note that we can think of optimizing the function $\tilde{B} \mapsto \left\|\tilde{B} - B\right\|_F^2$, which is convex on $\mathbb{R}^{n \times n}$. The constraint set

$$\{\tilde{B} \in \mathbb{R}^{n \times n} : \tilde{B}s = y\}$$

is affine and hence convex. Therefore, the optimization problem admits at most one minimizer.

We show that $\bar{B}_{\text{Broyden}}$ is indeed the minimizer. For any $\tilde{B}$ satisfying $\tilde{B}s = y$, we have

$$\begin{aligned}
\left\|\bar{B}_{\text{Broyden}} - B\right\|_F^2 &= \left\|\frac{(y - Bs)s^\top}{s^\top s}\right\|_F^2 && (\text{definition of } \bar{B}_{\text{Broyden}}) \\
&= \left\|(\tilde{B} - B)\frac{ss^\top}{s^\top s}\right\|_F^2 && (\tilde{B}s = y) \\
&\le \left\|\tilde{B} - B\right\|_F^2.
\end{aligned}$$

In the last inequality, we used the submultiplicativity of the Frobenius norm and the fact that $\left\|ss^\top/(s^\top s)\right\|_F = 1$. Hence $\tilde{B} = \bar{B}_{\text{Broyden}}$, and we obtain the claim. □

Note that this update rule does not preserve symmetry. In the following, we introduce representative quasi-Newton update rules that improve upon this aspect.

3.2.2 Symmetric Rank-One (SR1) Update

The Symmetric Rank-One (SR1) update [6] is a fundamental quasi-Newton method that maintains symmetry throughout the update process. The update formulas are given by

$$\bar{B}_{\mathrm{SR1}} := B + \frac{(y-Bs)(y-Bs)^\top}{(y-Bs)^\top s}.$$

Derivation (SR1) To derive the SR1 update, we construct the updated matrix $\bar{B}$ as a rank-one update. That is, we assume that for some vector $z \in \mathbb{R}^n$,

$$\bar{B}_{\mathrm{SR1}} = B + zz^\top.$$

To satisfy the secant condition $\bar{B}_{\mathrm{SR1}} s = y$, when $z^\top s \neq 0$, we need

$$Bs + zz^\top s = y.$$

Rearranging this equation gives

$$z = \frac{y-Bs}{z^\top s}.$$

To determine $z^\top s$, we take the inner product with s:

$$z^\top s = \frac{(y-Bs)^\top s}{z^\top s}.$$

Rearranging this equation yields the key relation

$$(z^\top s)^2 = (y-Bs)^\top s.$$

Thus, we obtain

$$\bar{B}_{\mathrm{SR1}} = B + zz^\top = B + \frac{(y-Bs)(y-Bs)^\top}{(z^\top s)^2} = B + \frac{(y-Bs)(y-Bs)^\top}{(y-Bs)^\top s}.$$

Thus, we have derived the SR1 update formula. We omit the derivation of $\bar{H}_{\mathrm{SR1}}$.

Remarks The SR1 update requires $(y-Bs)^\top s \neq 0$ to be well-defined. When $(y-Bs)^\top s = 0$, the denominator of the SR1 update vanishes, and the update is typically skipped in practice.

3.2.3 Powell Symmetric Broyden (PSB) Update

The Powell Symmetric Broyden (PSB) update [9–11] is also a quasi-Newton update rule. The update formulas are given by

$$\bar{B}_{\mathrm{PSB}} = B + \frac{(y-Bs)s^\top + s(y-Bs)^\top}{s^\top s} - \frac{s^\top(y-Bs)}{(s^\top s)^2} ss^\top$$

Derivation (PSB) Recall that the SR1 update adds $zz^\top$ to B to obtain $\bar{B}$. Instead, we can consider an asymmetric rank-one update of the form $zc^\top$ for some vector $c \in \mathbb{R}^n$. We symmetrize the result at the end. Assume $c^\top s \neq 0$, and from $Bs + zc^\top s = y$, we can derive z as

$$z = \frac{y-Bs}{c^\top s}$$

and perform the asymmetric update

$$C_1 := B + \frac{(y-Bs)c^\top}{c^\top s}.$$

Since C_1 is generally not symmetric, we symmetrize it via

$$C_2 = \frac{C_1 + C_1^\top}{2}.$$

However, the symmetrized matrix C_2 may not satisfy the secant condition $C_2 s = y$. Therefore, we iterate this process:

$$\begin{cases} C_0 = B \\ C_{2t+1} = C_{2t} + \dfrac{(y - C_{2t}s)c^\top}{c^\top s} & \text{(asymmetric update)} \\ C_{2t+2} = \dfrac{C_{2t+1} + C_{2t+1}^\top}{2} & \text{(symmetrization)} \end{cases}$$

The key result is that the sequence $\{C_t\}_t$ converges to a symmetric matrix satisfying the secant condition, as formalized in the following proposition.

Proposition 10 ([8, Lemma 7.2])**.** *Suppose that $c^\top s \neq 0$. Then, the sequence $\{C_t\}_t$ converges, and the limit is given by*

$$\lim_{t\to\infty} C_t = C_\infty := B + \frac{(y - Bs)c^\top + c(y - Bs)^\top}{c^\top s} - \frac{(y - Bs)^\top s}{(c^\top s)^2} cc^\top.$$

Proof. We first analyze the even subsequence. For $t = 0, 1, 2, \ldots$, we define

$$G_t := C_{2t}.$$

By construction, each G_t is symmetric. From the definitions, we have

$$G_{t+1} = G_t + \frac{1}{2c^\top s}\left((y - G_t s)c^\top + c(y - G_t s)^\top\right).$$

Let us introduce the error vector for the secant condition:

$$w_t := y - G_t s.$$

Then, we have

$$G_{t+1} = G_t + \frac{1}{2c^\top s}(w_t c^\top + c w_t^\top).$$

Substituting the above equation into the definition of w_t yields

$$\begin{aligned} w_{t+1} &= y - \left(G_t + \frac{1}{2c^\top s}(w_t c^\top + c w_t^\top)\right) s \\ &= w_t - \frac{1}{2} w_t - \frac{w_t^\top s}{2c^\top s} c \\ &= \frac{1}{2}\left(w_t - \frac{w_t^\top s}{c^\top s} c\right). \end{aligned}$$

Hence, we have

$$w_{t+1} = P w_t, \qquad P := \frac{1}{2}\left(I - \frac{cs^\top}{c^\top s}\right).$$

The matrix $cs^\top / c^\top s$ is rank one and has eigenvalues $1, 0, \ldots, 0$. Therefore, P has one eigenvalue 0 and all remaining eigenvalues equal to $1/2$. In particular, its spectral radius is $1/2 < 1$, and thus the series converges as follows:

$$\begin{aligned} \sum_{t=0}^{\infty} w_t &= \sum_{t=0}^{\infty} P^t (y - Bs) && (w_0 = y - Bs) \\ &= (I - P)^{-1}(y - Bs) \\ &= 2\left(I - \frac{1}{2}\frac{cs^\top}{c^\top s}\right)(y - Bs). && \text{(definition of } P\text{)} \end{aligned}$$

Note that the last equation follows from

$$2(I - P)\left(I - \frac{1}{2}\frac{cs^\top}{c^\top s}\right) = \left(I + \frac{cs^\top}{c^\top s}\right)\left(I - \frac{1}{2}\frac{cs^\top}{c^\top s}\right) = I.$$

Hence, we have

$$\begin{aligned}
\lim_{t\to\infty} G_t &= B + \frac{1}{2c^\top s}\sum_{t=0}^{\infty}(w_t c^\top + c w_t^\top) \\
&= B + \left(\sum_{t=0}^{\infty} w_t\right)\frac{c^\top}{2c^\top s} + \frac{c}{2c^\top s}\left(\sum_{t=0}^{\infty} w_t\right)^\top \\
&= B + 2\left(I - \frac{1}{2}\frac{cs^\top}{c^\top s}\right)(y-Bs)\frac{c^\top}{2c^\top s} + \frac{c}{2c^\top s}2(y-Bs)^\top\left(I - \frac{1}{2}\frac{sc^\top}{c^\top s}\right) \\
&= B + \frac{(y-Bs)c^\top + c(y-Bs)^\top}{c^\top s} - \frac{(y-Bs)^\top s}{(c^\top s)^2}cc^\top \\
&= C_\infty.
\end{aligned}$$

This shows the convergence of the even subsequence $\{C_{2t}\}_t$ to C_∞.

Next, for the odd subsequence, we have

$$C_{2t+1} = G_t + \frac{w_t c^\top}{c^\top s}.$$

Since $G_t \to C_\infty$ and $\|w_t\| \to 0$ from the spectral radius of P being less than one, we have

$$C_{2t+1} \to C_\infty.$$

Therefore both subsequences $\{C_{2t}\}_t$ and $\{C_{2t+1}\}_t$ converge to C_∞, completing the proof. □

When $c = s$, the general formula of C_∞ simplifies to the standard PSB update formula:

$$\bar{B}_{\mathrm{PSB}} = B + \frac{(y-Bs)s^\top + s(y-Bs)^\top}{s^\top s} - \frac{(y-Bs)^\top s}{(s^\top s)^2}ss^\top.$$

3.2.4 Davidon–Fletcher–Powell (DFP) Update

The Davidon–Fletcher–Powell (DFP) update [6] is also a classical quasi-Newton update formula. The update formulas are given by

$$\bar{B}_{\mathrm{DFP}} = \left(I - \frac{ys^\top}{y^\top s}\right)B\left(I - \frac{sy^\top}{y^\top s}\right) + \frac{yy^\top}{y^\top s}.$$

Derivation (DFP) In the PSB update rule discussed earlier, we substituted $c = s$, but we can consider taking a different c. Specifically, substituting $c = y$ yields the DFP update as follows.

$$\bar{B}_{\mathrm{DFP}} = B + \frac{(y-Bs)y^\top + y(y-Bs)^\top}{y^\top s} - \frac{(y-Bs)^\top s}{(y^\top s)^2}yy^\top.$$

There are other derivations of the DFP update, some of which can be understood as dual versions of the derivation of the subsequent BFGS update.

3.2.5 Broyden–Fletcher–Goldfarb–Shanno (BFGS) Update

The Broyden–Fletcher–Goldfarb–Shanno (BFGS) update is one of the most widely used quasi-Newton methods. The update formulas are given by

$$\bar{B}_{\mathrm{BFGS}} = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}.$$

Derivation by Duality This update can be derived by considering the dual of the DFP update. The secant condition is $\bar{B}s = y$, and assuming $\bar{B}$ is invertible, we can also write it as $\bar{H}y = s$. By performing the same operations of enforcing the secant condition and symmetrization on this condition, we obtain

$$\bar{H}_{\mathrm{BFGS}} = \left(I - \frac{sy^\top}{y^\top s}\right) H \left(I - \frac{ys^\top}{y^\top s}\right) + \frac{ss^\top}{y^\top s}.$$

As we will see later, this is indeed the inverse of $\bar{B}_{\mathrm{BFGS}}$. Note that this has the same form as the DFP update with (B, s, y) replaced by (H, y, s). Therefore, we can say that the BFGS update is the dual of the DFP update.

Derivation by KL Divergence Minimization The KL divergence is a non-symmetric measure of proximity between matrices. For positive definite matrices $P, Q \in \mathbb{R}^{n\times n}$, it is defined by

$$\mathrm{KL}(P, Q) := \mathrm{tr}(PQ^{-1}) - \log\det\bigl(PQ^{-1}\bigr) - n.$$

Let $T \in \mathbb{R}^{n\times n}$ be an arbitrary nonsingular matrix. Since the trace is invariant under cyclic permutations, the KL divergence satisfies

$$\begin{aligned}
\mathrm{KL}(TPT^\top, TQT^\top) &= \mathrm{tr}(TPT^\top(TQT^\top)^{-1}) - \log\det\Bigl(TPT^\top(TQT^\top)^{-1}\Bigr) - n \\
&= \mathrm{tr}(PQ^{-1}T^{-1}T) - \log\bigl(\det(T)\det\bigl(PQ^{-1}\bigr)\det\bigl(T^{-1}\bigr)\bigr) - n \\
&= \mathrm{tr}(PQ^{-1}) - \log\det\Bigl(PQ^{-1}\Bigr) - n \\
&= \mathrm{KL}(P, Q).
\end{aligned}$$

It is known that the BFGS update can be obtained as the solution to the following KL divergence minimization problem:

$$\begin{aligned}
\underset{\bar{B}\succ 0}{\text{minimize}} \quad & \mathrm{KL}(\bar{B}, B) \\
\text{subject to} \quad & \bar{B}s = y.
\end{aligned}$$

Proposition 11 ([2, Section 7.2.4])**.** *Let $y^\top s > 0$ and $B \succ 0$ be a symmetric positive definite matrix. The solution to the above optimization problem is given by the BFGS update formula.*

Proof. Since B is a symmetric positive definite matrix, we first define

$$s' := B^{\frac{1}{2}}s, \quad y' := B^{-\frac{1}{2}}y, \quad B' := B^{-\frac{1}{2}}\bar{B}B^{-\frac{1}{2}}.$$

Then, the optimization problem can be rewritten as

$$\begin{aligned}
\underset{B'\succ 0}{\text{minimize}} \quad & \mathrm{KL}(B', I) \\
\text{subject to} \quad & B's' = y'.
\end{aligned}$$

Here, we used the invariance of the KL divergence:

$$\mathrm{KL}(\bar{B}, B) = \mathrm{KL}(B^{-\frac{1}{2}}\bar{B}B^{-\frac{1}{2}}, B^{-\frac{1}{2}}BB^{-\frac{1}{2}}) = \mathrm{KL}(B', I)$$

Note that the constraint $B' \succ 0$ automatically includes symmetry, so adding the constraint $B' = B'^\top$ explicitly does not change the solution.

Let us solve this problem using the method of Lagrange multipliers. We introduce the Lagrange multipliers $\lambda \in \mathbb{R}^n$ and $\Lambda \in \mathbb{R}^{n\times n}$, and define the Lagrangian as

$$\mathscr{L}(B', \lambda, \Lambda) = \mathrm{tr}(B') - \log\det\bigl(B'\bigr) - n + 2\lambda^\top(B's' - y') + \mathrm{tr}\left(\Lambda(B' - B'^\top)\right).$$

In general, for matrix differentiation, the derivative of $\log\det(X)$ with respect to X for $X \succ 0$ is $X^{-\top}$, and the derivative of $\mathrm{tr}(AX)$ is $A^\top$. By differentiating with respect to B' and using $B' = B'^\top$, we can write down the stationarity condition of the KKT conditions as follows:

$$I - B'^{-\top} + 2\lambda s'^\top + \Lambda^\top - \Lambda = 0.$$

By adding this equation and its transpose, and dividing by 2 again using $B' = B'^{\top}$, we obtain

$$B'^{-1} = I + \lambda s'^{\top} + s'\lambda^{\top}.$$

By the constraint $B's' = y'$, we have $B'^{-1}y' = s'$, so multiplying the above equation by y' from the right and then by $y'^{\top}$ from the left yields

$$\begin{aligned} s' &= y' + (s'^{\top}y')\lambda + (\lambda^{\top}y')s', \\ y'^{\top}s' &= y'^{\top}y' + (s'^{\top}y')(y'^{\top}\lambda) + (\lambda^{\top}y')(y'^{\top}s'). \end{aligned}$$

From the second equation, we have

$$\lambda^{\top}y' = \frac{y'^{\top}s' - y'^{\top}y'}{2(s'^{\top}y')}.$$

Substituting this into the first equation gives

$$\lambda = \frac{s' - y'}{s'^{\top}y'} - \frac{\lambda^{\top}y'}{s'^{\top}y'}s' = \frac{s'^{\top}y' + y'^{\top}y'}{2(s'^{\top}y')^2}s' - \frac{1}{s'^{\top}y'}y'.$$

By substituting this λ into the expression for B'^{-1} and simplifying, we obtain

$$\begin{aligned} B'^{-1} &= I + 2\frac{s'^{\top}y' + y'^{\top}y'}{2(s'^{\top}y')^2}(s's'^{\top}) - \frac{1}{s'^{\top}y'}(y's'^{\top} + s'y'^{\top}) \\ &= \left(I - \frac{s'y'^{\top}}{s'^{\top}y'}\right)\left(I - \frac{y's'^{\top}}{s'^{\top}y'}\right) + \frac{s's'^{\top}}{s'^{\top}y'}. \end{aligned}$$

Finally, since $B' = B^{-\frac{1}{2}}\bar{B}B^{-\frac{1}{2}}$, we have $\bar{B} = B^{\frac{1}{2}}B'B^{\frac{1}{2}}$, and thus calculating $\bar{B}^{-1}$ gives

$$\begin{aligned} \bar{B}^{-1} &= B^{-\frac{1}{2}}B'^{-1}B^{-\frac{1}{2}} \\ &= B^{-\frac{1}{2}}\left(I - \frac{s'y'^{\top}}{s'^{\top}y'}\right)\left(I - \frac{y's'^{\top}}{s'^{\top}y'}\right)B^{-\frac{1}{2}} + \frac{(B^{-\frac{1}{2}}s')(B^{-\frac{1}{2}}s')^{\top}}{s'^{\top}y'} \\ &= B^{-\frac{1}{2}}\left(I - \frac{B^{\frac{1}{2}}sy^{\top}B^{-\frac{1}{2}}}{s^{\top}y}\right)\left(I - \frac{B^{-\frac{1}{2}}ys^{\top}B^{\frac{1}{2}}}{s^{\top}y}\right)B^{-\frac{1}{2}} + \frac{ss^{\top}}{s^{\top}y} \\ &= \left(I - \frac{sy^{\top}}{y^{\top}s}\right)B^{-1}\left(I - \frac{ys^{\top}}{y^{\top}s}\right) + \frac{ss^{\top}}{y^{\top}s}. \end{aligned}$$

As we will see later, this is indeed the inverse of the BFGS update and positive definite, the proof is complete. ☐

As for more details, see [12, 13].

3.3 Details of BFGS Update

In this subsection, we focus on the BFGS update and provide a detailed derivation of its formula. BFGS update is known as one of the most successful quasi-Newton update rules in practice. Recall that the BFGS update formula is given by

$$B_{k+1} = B_k - \frac{B_k s_k s_k^{\top} B_k}{s_k^{\top} B_k s_k} + \frac{y_k y_k^{\top}}{y_k^{\top} s_k}.$$

3.3.1 Formula for the Inverse Update

The inverse of the BFGS update $H_k := B_k^{-1}$ is given by

$$H_{k+1} = \left(I - \frac{s_k y_k^{\top}}{y_k^{\top} s_k}\right) H_k \left(I - \frac{y_k s_k^{\top}}{y_k^{\top} s_k}\right) + \frac{s_k s_k^{\top}}{y_k^{\top} s_k}.$$

Note that this BFGS update formula for H_{k+1} has the same form as the DFP update formula for B_{k+1}. These are dual to each other. We now prove that this equation indeed gives the inverse of the BFGS update.

Proposition 12. *The matrix H_{k+1} is the inverse of B_{k+1}.*

Proof. The BFGS update can be rewritten in a compact rank-two form

$$B_{k+1} = B_k + UCV^\top.$$

Here, we define

$$U = \begin{bmatrix} B_k s_k & y_k \end{bmatrix}, \qquad C = \begin{pmatrix} -\frac{1}{s_k^\top B_k s_k} & 0 \\ 0 & \frac{1}{y_k^\top s_k} \end{pmatrix}, \qquad V = \begin{bmatrix} B_k s_k & y_k \end{bmatrix}$$

since

$$UCV^\top = \begin{bmatrix} -\frac{B_k s_k}{s_k^\top B_k s_k} & \frac{y_k}{y_k^\top s_k} \end{bmatrix} \begin{bmatrix} s_k^\top B_k \\ y_k^\top \end{bmatrix} = -\frac{B_k s_k s_k^\top B_k}{s_k^\top B_k s_k} + \frac{y_k y_k^\top}{y_k^\top s_k}.$$

By the Sherman–Morrison–Woodbury identity, we can compute H_{k+1} as follows:

$$\begin{aligned}
H_{k+1} &= (B_k + UCV^\top)^{-1} \\
&= B_k^{-1} - B_k^{-1} U \left(C^{-1} + V^\top B_k^{-1} U \right)^{-1} V^\top B_k^{-1} \\
&= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \left(\begin{pmatrix} -s_k^\top B_k s_k & 0 \\ 0 & y_k^\top s_k \end{pmatrix} + \begin{pmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top H_k y_k \end{pmatrix} \right)^{-1} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\
&= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \begin{pmatrix} 0 & y_k^\top s_k \\ y_k^\top s_k & y_k^\top H_k y_k + y_k^\top s_k \end{pmatrix}^{-1} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\
&= H_k - \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \left(-\frac{1}{(y_k^\top s_k)^2} \begin{pmatrix} y_k^\top H_k y_k + y_k^\top s_k & -y_k^\top s_k \\ -y_k^\top s_k & 0 \end{pmatrix} \right) \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\
&= H_k + \frac{1}{(y_k^\top s_k)^2} \begin{bmatrix} s_k & H_k y_k \end{bmatrix} \begin{pmatrix} y_k^\top H_k y_k + y_k^\top s_k & -y_k^\top s_k \\ -y_k^\top s_k & 0 \end{pmatrix} \begin{bmatrix} s_k^\top \\ y_k^\top H_k \end{bmatrix} \\
&= H_k + \frac{1}{(y_k^\top s_k)^2} \left((y_k^\top H_k y_k + y_k^\top s_k) s_k s_k^\top - (y_k^\top s_k)(s_k y_k^\top H_k + H_k y_k s_k^\top) \right) \\
&= \left(I - \frac{s_k y_k^\top}{y_k^\top s_k} \right) H_k \left(I - \frac{y_k s_k^\top}{y_k^\top s_k} \right) + \frac{s_k s_k^\top}{y_k^\top s_k}.
\end{aligned}$$

which completes the proof. □

3.3.2 Positive Definiteness of the BFGS Update

An important property of the BFGS update is that if the current approximation B_k is positive definite and the curvature condition $y_k^\top s_k > 0$ holds, then the updated approximation B_{k+1} is also guaranteed to be positive definite.

Proposition 13. *If B_k is positive definite and $y_k^\top s_k > 0$, then B_{k+1} is also positive definite.*

Proof. By assumption, B_k and its inverse H_k are positive definite. For any nonzero vector $v \in \mathbb{R}^n$, we have

$$v^\top H_{k+1} v = v^\top \left(I - \frac{s_k y_k^\top}{y_k^\top s_k} \right) H_k \left(I - \frac{y_k s_k^\top}{y_k^\top s_k} \right) v + v^\top \frac{s_k s_k^\top}{y_k^\top s_k} v \geq 0 + \frac{(s_k^\top v)^2}{y_k^\top s_k} > 0.$$

The first term is nonnegative since H_k is positive definite, and the second term is positive due to the curvature condition $y_k^\top s_k > 0$. Thus, H_{k+1} is positive definite, and consequently, $B_{k+1} = H_{k+1}^{-1}$ is also positive definite. □

This proposition is crucial for ensuring the positive definiteness of the approximate Hessian matrices B_k throughout the iterations, which in turn guarantees that the search directions are descent directions. In practice, it is common to use a line search that satisfies the following strong curvature condition, included in the strong Wolfe conditions:

$$|g_{k+1}^\top d_k| \leq c_2 |g_k^\top d_k|.$$

Here, $0 < c_2 < 1$ is a constant. By imposing this condition, we can verify that B_k remains positive definite throughout the iterations. First, we initialize B_0 to be positive definite. Assuming that B_k is positive definite for some k, since d_k is a descent direction with $g_k^\top d_k < 0$, and from $s_k = \alpha_k d_k$ and $y_k = g_{k+1} - g_k$, we have

$$\begin{aligned}
y_k^\top s_k &= (g_{k+1} - g_k)^\top (\alpha_k d_k) \\
&= \alpha_k (g_{k+1}^\top d_k - g_k^\top d_k) \\
&\geq \alpha_k \left(-|g_{k+1}^\top d_k| + |g_k^\top d_k| \right) && (g_k^\top d_k < 0) \\
&\geq \alpha_k (1 - c_2) |g_k^\top d_k| && \text{(by the curvature condition)} \\
&> 0. && (c_2 < 1)
\end{aligned}$$

Therefore, the curvature condition $y_k^\top s_k > 0$ is satisfied, and by the previous proposition, B_{k+1} is also positive definite. Thus, by mathematical induction, all B_k are guaranteed to be positive definite for every k.

3.3.3 Trace and Determinant Formulas for the BFGS Update

We also present trace and determinant formulas for the BFGS update, which are useful in analyzing the eigenvalue behavior of the updated matrix. For symmetric matrices, these quantities correspond to the sum and product of eigenvalues, respectively. Hence, if both the trace and the determinant are appropriately bounded, one may expect the eigenvalues themselves to remain bounded. This is closely related to assumptions on the Hessian eigenvalues of the objective function, such as μ-strong convexity and L-smoothness.

Trace Formula The trace of the BFGS-updated matrix has an explicit formula.

Proposition 14 ([6, eq. 6.44])**.** *Let $B_+ = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}$ be the BFGS update. Then*

$$\operatorname{tr}(B_+) = \operatorname{tr}(B) - \frac{\|Bs\|^2}{s^\top Bs} + \frac{\|y\|^2}{y^\top s}.$$

Proof. Applying the trace to the BFGS update formula, we have

$$\operatorname{tr}(B_+) = \operatorname{tr}(B) - \operatorname{tr}\left(\frac{Bss^\top B}{s^\top Bs} \right) + \operatorname{tr}\left(\frac{yy^\top}{y^\top s} \right).$$

For the second term, note that

$$\operatorname{tr}\left(\frac{Bss^\top B}{s^\top Bs} \right) = \frac{1}{s^\top Bs} \operatorname{tr}((Bs)(Bs)^\top) = \frac{\|Bs\|^2}{s^\top Bs}.$$

For the third term, we have

$$\operatorname{tr}\left(\frac{yy^\top}{y^\top s} \right) = \frac{1}{y^\top s} \operatorname{tr}(yy^\top) = \frac{\|y\|^2}{y^\top s}.$$

Combining these results yields the desired formula. □

Determinant Formula The determinant of the BFGS-updated matrix also admits a closed-form expression.

Proposition 15 ([6, eq. 6.45])**.** *Let $B_+ = B - \frac{Bss^\top B}{s^\top Bs} + \frac{yy^\top}{y^\top s}$ be the BFGS update and assume that B is nonsingular. Then*

$$\det(B_+) = \det(B) \frac{y^\top s}{s^\top Bs}.$$

Proof. Recall the rank-two formulation of the BFGS update. By the matrix determinant lemma, the matrices U, C, V satisfy

$$\det(B_{k+1}) = \det\Big(B_k + UCV^\top\Big) = \det(B_k) \det(C) \det\left(C^{-1} + V^\top B_k^{-1} U\right).$$

Since $U = V = \begin{bmatrix} B_k s_k & y_k \end{bmatrix}$, we have

$$V^\top B_k^{-1} U = \begin{bmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top B_k^{-1} y_k \end{bmatrix},$$

and thus

$$C^{-1} + V^\top B_k^{-1} U = \begin{bmatrix} -s_k^\top B_k s_k & 0 \\ 0 & y_k^\top s_k \end{bmatrix} + \begin{bmatrix} s_k^\top B_k s_k & s_k^\top y_k \\ y_k^\top s_k & y_k^\top B_k^{-1} y_k \end{bmatrix} = \begin{bmatrix} 0 & s_k^\top y_k \\ y_k^\top s_k & y_k^\top s_k + y_k^\top B_k^{-1} y_k \end{bmatrix}.$$

Thus, combining these results, we obtain

$$\det(B_{k+1}) = \det(B_k) \left(-\frac{1}{s_k^\top B_k s_k} \cdot \frac{1}{y_k^\top s_k} \right) \left(-(s_k^\top y_k)(y_k^\top s_k) \right) = \det(B_k) \frac{y_k^\top s_k}{s_k^\top B_k s_k},$$

which completes the proof. □

3.4 BFGS vs DFP

As mentioned several times earlier, although BFGS and DFP are dual to each other, they exhibit remarkably different practical efficiency when applied to real optimization problems. Powell's analysis [14] investigates this asymmetry by studying the behavior of both methods on a simple two-dimensional quadratic function. While asymptotic convergence theory often suggests that both methods should perform similarly, Powell reveals that their practical efficiency can differ dramatically, especially when the approximate Hessian is far from the true Hessian. Here, we reproduce his numerical experiments.

3.4.1 Problem Setup

Following Powell's framework, we consider the quadratic function

$$f(x,y) = \frac{1}{2}(x^2 + y^2).$$

The key aspect of this experiment is the choice of the initial Hessian approximation B_0. In principle, the true Hessian of this function is the identity matrix, but here we deliberately use a significantly incorrect initial approximation to test the robustness and corrective ability of the methods against such errors. Specifically, we choose the initial Hessian approximation B_0 to have eigenvalues 1 and λ_1, where λ_1 represents the degree of error in the initial approximation. Both BFGS and DFP methods used a fixed step size of $\alpha_k = 1$ at every iteration.

Additionally, we chose the initial point x_0 as follows:

$$\theta = \arctan\left(\sqrt{\lambda_1}\right), \quad x_0 = \begin{bmatrix} \cos(\theta) \\ \sin(\theta) \end{bmatrix}.$$

This aligns with Powell's original analysis. See [14] for details on this choice.

The iteration continues until the norm of the current point falls below a tolerance relative to its initial norm. For each value of λ_1, we record the number of iterations required for convergence.

3.4.2 Experimental Results

The numerical results are presented in Table 1, which partially reproduces the content of Powell's original tables. We can see that the convergence behavior depends critically on the initial eigenvalue λ_1.

Figure 11 and Figure 12 illustrate the iterative trajectories for specific values of λ_1. These figures visualize how the two methods navigate toward the minimum (at the origin) from the same initial point, clearly showing the difference in convergence speed and path.

3.4.3 Analysis and Discussion

The numerical results reveal a striking asymmetry between BFGS and DFP. When $\lambda_1 > 1$ (i.e., the initial Hessian approximation overestimates the true Hessian curvature), BFGS demonstrates significantly better efficiency. Conversely, when $\lambda_1 < 1$ (i.e., the initial Hessian approximation underestimates the true curvature), DFP performs slightly better than BFGS, though the difference is modest. This trend reversal can be predicted by the theoretical symmetry between the two methods, but the asymmetry in the magnitude is noteworthy.

λ_1	BFGS	DFP
0.001	4	3
0.01	5	3
0.1	6	4
1	1	1
10	8	16
100	10	107
1000	12	1006
10000	15	9987

Table 1: Convergence comparison between BFGS and DFP methods for different initial eigenvalues λ_1.

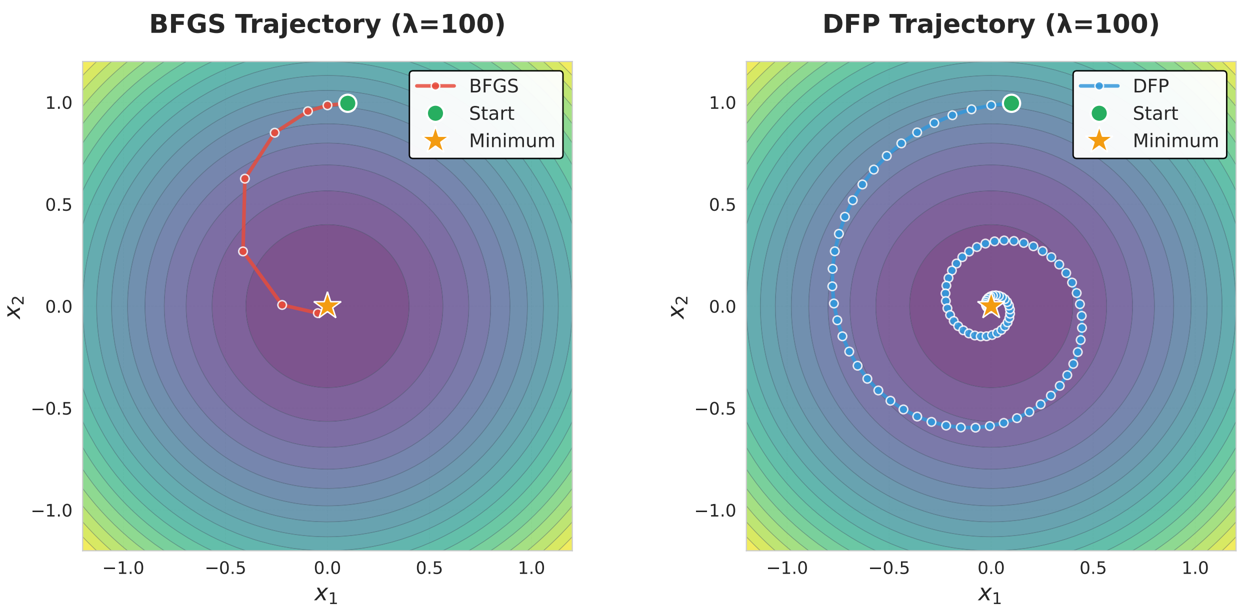

Figure 11: Iterative trajectories for BFGS and DFP methods when $\lambda_1 = 100$. BFGS converges in 10 iterations while DFP requires 107 iterations, showing BFGS's superior efficiency for large eigenvalue errors.

Asymmetry in Hessian Correction The asymmetry of the performance arises from a fundamental difference in how these two methods correct erroneous eigenvalues. The core insight is that correcting erroneously large eigenvalues is more critical than correcting erroneously small ones.

When a Hessian eigenvalue is overestimated, the algorithm takes steps that are too conservative, resulting in slow progress of the optimization. Correcting such errors requires the update formula to reduce these large eigenvalues toward one. The BFGS update is highly effective at this task, while the DFP update struggles.

In contrast, when a Hessian eigenvalue is underestimated, the algorithm takes steps that are slightly too aggressive, but the error is self-correcting. Subsequent gradient computations provide information that helps refine the approximation. Thus, correcting underestimated eigenvalues is inherently easier and requires fewer iterations.

This explains why the asymmetry in performance exists between BFGS and DFP.

3.5 Limited Memory BFGS (L-BFGS)

Next, we discuss the limited memory quasi-Newton methods, which are crucial for extending quasi-Newton methods to large-scale optimization problems. In standard quasi-Newton methods, the approximate Hessian B_k or its inverse H_k is stored and updated explicitly as a dense matrix, requiring $\mathscr{O}(n^2)$ memory for n variables. In contrast, limited memory quasi-Newton methods maintain only a limited amount of information, i.e., the most recent m curvature pairs $\{(s_i, y_i)\}$ and perform computations related to the approximate Hessian using just this information. This reduces the space complexity to $\mathscr{O}(nm)$, which is a dramatic improvement when m is a small constant (typically $m \leq 10$).

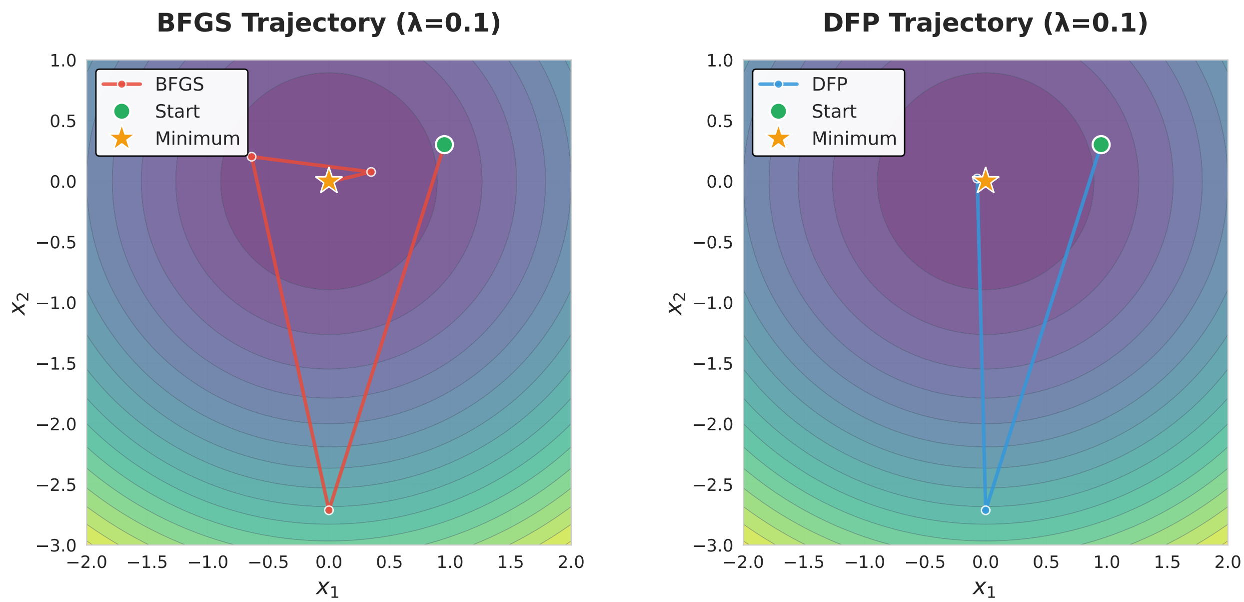

Figure 12: Iterative trajectories for BFGS and DFP methods when $\lambda_1 = 0.1$. Both methods converge quickly, with DFP slightly faster than BFGS.

The L-BFGS method [15], which is a limited memory version of the BFGS update, is particularly representative. We focus on this BFGS update case and show how the quasi-Newton direction $d_k = -H_k g_k$ can be computed efficiently in terms of both space and time complexity using only the limited information.

Throughout this subsection, we work with a finite sequence of matrices.

$$H_0, H_1, \ldots, H_m.$$

Here, H_ℓ denotes the inverse Hessian approximation obtained after ℓ BFGS updates applied to a given initial matrix H_0. This is done for the sake of clarity in the discussion, but it can also be thought of as maintaining a rolling window of the most recent m inverse Hessian approximations corresponding to the past m BFGS updates when the current iteration count is k.

3.5.1 Compact Representation of the Inverse Hessian

Let $\{(s_i, y_i)\}_{i=0}^{m-1}$ be the stored correction pairs, and define

$$\rho_i = \frac{1}{y_i^\top s_i}, \qquad V_i = I - \rho_i y_i s_i^\top.$$

Then, for $i = 0, \ldots, m-1$, the inverse BFGS update can be expressed as

$$H_{i+1} = V_i^\top H_i V_i + \rho_i s_i s_i^\top.$$

By recursively expanding this relation, we obtain the compact representation

$$H_m = V_{m-1}^\top \cdots V_0^\top H_0 V_0 \cdots V_{m-1} + \left(\sum_{j=0}^{m-2} (V_{m-1}^\top \cdots V_{j+1}^\top) \rho_j s_j s_j^\top (V_{j+1} \cdots V_{m-1}) \right) + \rho_{m-1} s_{m-1} s_{m-1}^\top.$$

H_0 denotes the chosen initial inverse Hessian approximation, typically a scaled identity matrix.

3.5.2 Two-Loop Recursion

In quasi-Newton methods, it is crucial for the algorithm to efficiently compute $r = H_m q$ for the inverse matrix H_m and a given vector $q \in \mathbb{R}^n$. This operation can be carried out using two short loops of length m, known as the L-BFGS two-loop recursion [6, Algorithm 7.4]. This algorithm requires $\mathcal{O}(nm)$ time and space complexity.

Algorithm 1: L-BFGS Two-Loop Recursion for $r = H_m q$

Input: q, stored pairs $\{(s_i, y_i)\}_{i=0}^{m-1}$, initial matrix H_0
Output: $r = H_m q$
for $i = m-1, m-2, \ldots, 0$ **do**
  $\rho_i \leftarrow 1/(y_i^\top s_i)$
  $\alpha_i \leftarrow \rho_i s_i^\top q$
  $q \leftarrow q - \alpha_i y_i$
end
$r \leftarrow H_0 q$
for $i = 0, \ldots, m-1$ **do**
  $\beta_i \leftarrow \rho_i y_i^\top r$
  $r \leftarrow r + s_i(\alpha_i - \beta_i)$
end
return r

In the following, we verify that the output of this two-loop recursion algorithm indeed computes $r = H_m q$.

Proposition 16. *The output of the two-loop recursion algorithm satisfies $r = H_m q$.*

Proof. In the first loop of the algorithm ($i = m-1, m-2, \ldots, 0$), from the input vector $q^{(m)} := q$, the algorithm computes

$$\alpha_i := \rho_i s_i^\top q^{(i+1)}, \qquad q^{(i)} := q^{(i+1)} - \alpha_i y_i.$$

Substituting the definition of α_i and using the definition of V_i in the compact representation, we find

$$q^{(i)} = q^{(i+1)} - \rho_i \left(s_i^\top q^{(i+1)} \right) y_i = \left(I - \rho_i y_i s_i^\top \right) q^{(i+1)} = V_i q^{(i+1)}.$$

Thus, for all $i = 0, 1, \ldots, m-1$, we have

$$q^{(i)} = V_i V_{i+1} \cdots V_{m-1} q.$$

Then, the algorithm applies H_0:

$$r^{(0)} = H_0 q^{(0)} = H_0 V_0 V_1 \cdots V_{m-1} q.$$

Next, for $i = 0, 1, \ldots, m-1$, the second loop computes

$$\beta_i = \rho_i y_i^\top r^{(i)}, \qquad r^{(i+1)} = r^{(i)} + s_i \left(\alpha_i - \beta_i \right).$$

Substituting the definitions of α_i, β_i, and $q^{(i+1)}$, we find that when $i < m-1$,

$$\begin{aligned}
r^{(i+1)} &= r^{(i)} + s_i \left(\rho_i s_i^\top q^{(i+1)} - \rho_i y_i^\top r^{(i)} \right) \\
&= r^{(i)} + \rho_i s_i s_i^\top \left(V_{i+1} V_{i+2} \cdots V_{m-1} \right) q - \rho_i s_i y_i^\top r^{(i)} \\
&= \left(I - \rho_i y_i s_i^\top \right) r^{(i)} + \rho_i s_i s_i^\top \left(V_{i+1} V_{i+2} \cdots V_{m-1} \right) q \\
&= V_i^\top r^{(i)} + \rho_i s_i s_i^\top \left(V_{i+1} V_{i+2} \cdots V_{m-1} \right) q,
\end{aligned}$$

and when $i = m-1$,

$$r^{(m)} = V_{m-1}^\top r^{(m-1)} + \rho_{m-1} s_{m-1} s_{m-1}^\top q.$$

By recursively expanding this relation from the initial value $r^{(0)} = H_0 q^{(0)}$, we obtain

$$r^{(m)} = V_{m-1}^\top \cdots V_0^\top H_0 V_0 \cdots V_{m-1} q + \left(\sum_{j=0}^{m-2} (V_{m-1}^\top \cdots V_{j+1}^\top) \rho_j s_j s_j^\top (V_{j+1} \cdots V_{m-1}) q \right) + \rho_{m-1} s_{m-1} s_{m-1}^\top q.$$

This matches the compact representation of H_m applied to q, and thus we have shown that the output of the algorithm satisfies $r = H_m q$, completing the proof. $\square$

Thus, the two-loop recursion exactly evaluates the action of the matrix H_m on a vector without explicitly constructing the matrix H_m.

3.5.3 Initial Step Size

A crucial component of the L-BFGS method is the choice of the initial matrix H_0. A widely used option is a scaled identity:

$$H_0 = \gamma I.$$

Here, the scaling parameter γ is often chosen as

$$\gamma = \frac{s_{m-1}^\top y_{m-1}}{y_{m-1}^\top y_{m-1}}.$$

Note that the pair (s_{m-1}, y_{m-1}) is the most recent one. This choice is motivated by the following discussion [15, 16]. Assume that the objective function f is twice continuously differentiable and consider the mean Hessian along the most recent step:

$$\bar{G} = \int_0^1 \nabla^2 f(x + \tau s_{m-1}) \mathrm{d}\tau.$$

Recall that s_{m-1} denotes the most recent displacement of x, satisfying

$$y_{m-1} = \nabla f(x + s_{m-1}) - \nabla f(x) = \bar{G} s_{m-1}.$$

Using this relation, the scaling factor can be written as

$$\frac{s_{m-1}^\top y_{m-1}}{y_{m-1}^\top y_{m-1}} = \frac{(\bar{G}^{1/2} s_{m-1})^\top (\bar{G}^{1/2} s_{m-1})}{(\bar{G}^{1/2} s_{m-1})^\top \bar{G} (\bar{G}^{1/2} s_{m-1})}.$$

This is a reciprocal of a Rayleigh quotient of the matrix $\bar{G}$ with respect to the vector $\bar{G}^{1/2} s_{m-1}$. Therefore, this scaling roughly approximates the average inverse curvature along those directions.

Moreover, this choice of γ coincides with the short step size of the Barzilai–Borwein method [17], highlighting a close connection between L-BFGS initialization and classical step-length selection strategies.

4 Modified Secant Condition

This section explains the modified secant condition, a modification of the standard secant condition. The modified secant condition incorporates function value information in addition to gradient information, leading to more accurate Hessian approximations.

4.1 Derivation of the Modified Secant Condition

Although the secant condition is widely used in many quasi-Newton methods, it can fail to capture the curvature of the objective function accurately because it only utilizes gradient information. We illustrate this limitation in Figure 13. In Figure 13, we have two points x_k and x_{k+1} where function values and gradients are already known. We see that the ideal quadratic model constructed from the exact Hessian fits well around the new point x_{k+1}, leading to a better convergence behavior. However, the standard secant condition, which only uses gradient information, ignores function value information and may misestimate curvature significantly, resulting in a poor approximation of the true objective function.

We can overcome this limitation by utilizing the function values. The basic idea is illustrated in Figure 14. Even when the gradients are the same at two points x_k and x_{k+1}, natural interpolation of function values differs depending on the function values $f(x_k)$ and $f(x_{k+1})$. This observation motivates the modified secant condition, which incorporates function value information to improve Hessian approximation.

In the following, we present two well-known modified secant conditions.

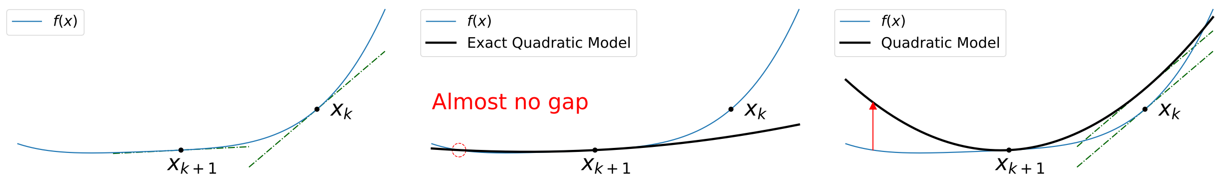

Figure 13: Limitations of the standard secant condition. (a) Function values and gradients at x_k and x_{k+1} are already known. (b) The ideal quadratic model constructed from the exact Hessian fits well around the new point x_{k+1}. (c) The standard secant condition may fail to capture curvature adequately.

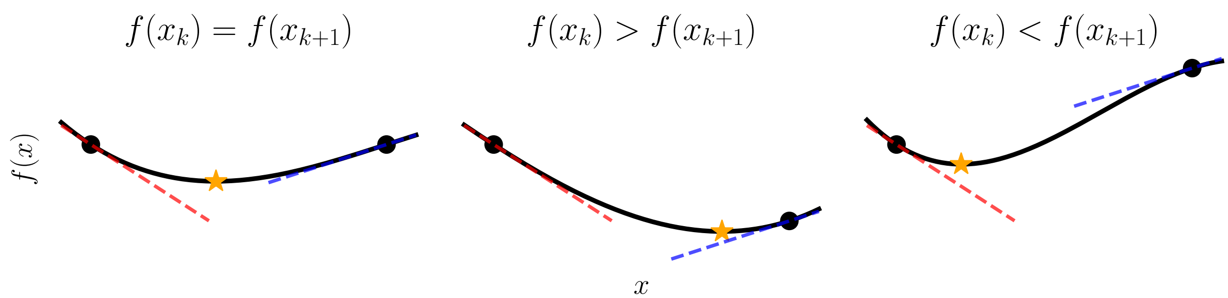

Figure 14: Basic idea of the modified secant condition. Even when the gradients are the same at two points x_k and x_{k+1}, natural interpolation of function values differs depending on the function values $f(x_k)$ and $f(x_{k+1})$.

4.1.1 Function-Value-Matching Modified Secant Condition

The first modification enforces the quadratic model to match the function value at the previous point [18–20]. Let us consider an alternate model with a different approximate Hessian B^{F}_{k+1}:

$$m^{\mathrm{F}}_{k+1}(x) := f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B^{\mathrm{F}}_{k+1}(x - x_{k+1}).$$

We require this model to satisfy

$$m^{\mathrm{F}}_{k+1}(x_k) = f(x_k),$$

which enforces that the function value at the previous point is correctly modeled.

Let us derive the corresponding modified secant condition for B^{F}_{k+1}. Substituting the definition of m^{F}_{k+1} and using $s_k = x_{k+1} - x_k$, the condition becomes

$$\begin{aligned} f(x_k) &= f(x_{k+1}) + \nabla f(x_{k+1})^\top (x_k - x_{k+1}) + \frac{1}{2}(x_k - x_{k+1})^\top B^{\mathrm{F}}_{k+1}(x_k - x_{k+1}) \\ &= f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2} s_k^\top B^{\mathrm{F}}_{k+1} s_k. \end{aligned}$$

Now, instead of the standard secant condition, we assume a modified form with an additional term involving the identity matrix:

$$\left(B^{\mathrm{F}}_{k+1} + \sigma^{\mathrm{F}}_k I\right) s_k = y_k.$$

Here, $\sigma^{\mathrm{F}}_k \in \mathbb{R}$ is a scalar to be determined as follows. Substituting this equation yields

$$f(x_k) = f(x_{k+1}) - \nabla f(x_{k+1})^\top s_k + \frac{1}{2} s_k^\top y_k - \frac{\sigma^{\mathrm{F}}_k}{2} \|s_k\|^2.$$

Solving for σ^{F}_k and using $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, we obtain

$$\sigma^{\mathrm{F}}_k = \frac{2(f(x_{k+1}) - f(x_k)) - (2\nabla f(x_{k+1}) - y_k)^\top s_k}{\|s_k\|^2}$$

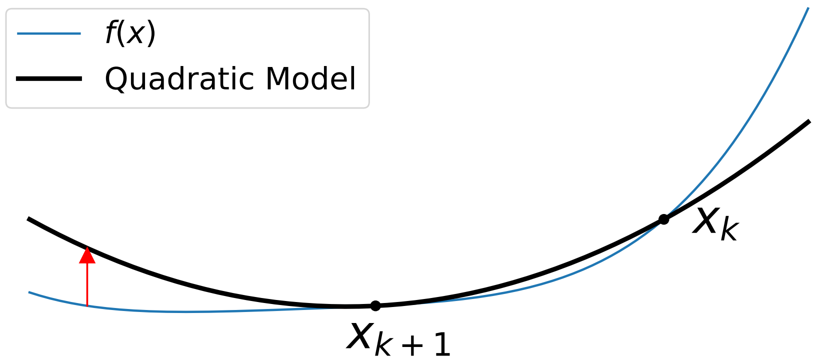

Figure 15: Function-value-matching modified secant equation. The function values $f(x_{k-1})$ and $m_k^{\mathrm{F}}(x_{k-1})$ are matched at the previous point x_{k-1}.

$$= \frac{2(f(x_{k+1}) - f(x_k)) - (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2}.$$

Therefore, the modified secant condition becomes

$$B_{k+1}^{\mathrm{F}} s_k = \hat{y}_k^{\mathrm{F}} := y_k + \frac{2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k}{\|s_k\|^2} s_k.$$

By using this modified $\hat{y}_k^{\mathrm{F}}$ instead of y_k in various quasi-Newton updates, we can incorporate function value information while keeping almost the same algorithm. Additionally, since condition $s_k^\top y_k > 0$ is important for maintaining positive definiteness in BFGS update, we can consider the following modification to ensure that the inner product with s_k remains positive:

$$B_{k+1}^{\mathrm{F}'} s_k = y_k + \frac{\max(0, 2(f(x_k) - f(x_{k+1})) + (\nabla f(x_{k+1}) + \nabla f(x_k))^\top s_k)}{\|s_k\|^2} s_k.$$

This is the function-value-matching modified secant condition. Refer to Figure 15 for an illustration.

4.1.2 Cubic-Augmented Modified Secant Condition

The second modification secant condition introduces a cubic term to the model, allowing simultaneous satisfaction of both the function value and the gradient matching at the previous point [21–23]. Let $T_{k+1} \in \mathbb{R}^{n \times n \times n}$ be the third-order derivative tensor of f at x_{k+1} such that

$$s_k^\top (T_{k+1} s_k) s_k = \sum_{i,j,l=1}^{n} \partial_{x_i x_j x_l} f(x_{k+1}) s_k^{(i)} s_k^{(j)} s_k^{(l)}.$$

Here, $\partial_{x_i x_j x_l} f$ denotes the third derivative of f with respect to x_i, x_j, and x_l, and $s_k^{(i)}$ is the i-th component of the vector s_k. We introduced this tensor solely for analysis purposes, and it will be eliminated from the final formula.

By incorporating this tensor term, we can define the following model:

$$\begin{aligned} m_{k+1}^{\mathrm{C}}(x) := f(x_{k+1}) + \nabla f(x_{k+1})^\top (x - x_{k+1}) + \frac{1}{2}(x - x_{k+1})^\top B_{k+1}^{\mathrm{C}} (x - x_{k+1}) \\ + \frac{1}{6}(x - x_{k+1})^\top (T_{k+1}(x - x_{k+1}))(x - x_{k+1}). \end{aligned}$$

With this model, we can enforce both function value and gradient matching at the previous point x_k. Specifically, we require

$$\begin{cases} m_{k+1}^{\mathrm{C}}(x_k) = f(x_k), \\ \nabla m_{k+1}^{\mathrm{C}}(x_k) = \nabla f(x_k). \end{cases}$$

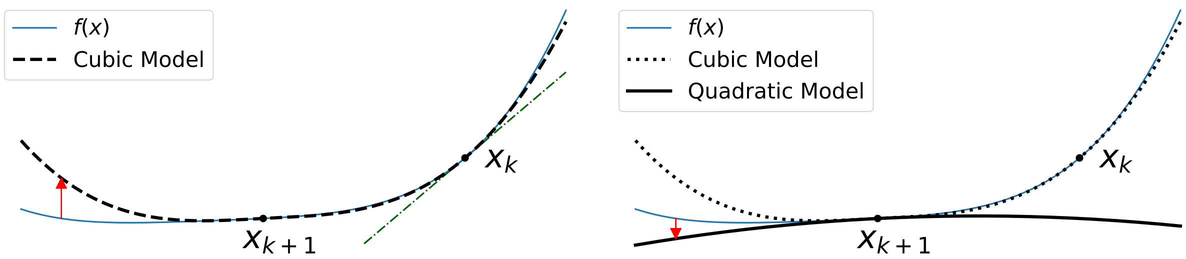

Figure 16: Cubic-augmented modified secant equation. The cubic term is incorporated into the model to simultaneously match both function value and gradient at the previous point. The model is constructed using the expansion up to the quadratic term.

By substituting the definition of m_{k+1}^{C} and using $s_k = x_{k+1} - x_k$, we can rewrite these conditions as

$$f(x_k) = f(x_{k+1}) - s_k^\top \nabla f(x_{k+1}) + \frac{1}{2} s_k^\top B_{k+1}^{\mathrm{C}} s_k - \frac{1}{6} s_k^\top (T_{k+1} s_k) s_k,$$
$$\nabla f(x_k) = \nabla f(x_{k+1}) - B_{k+1}^{\mathrm{C}} s_k + \frac{1}{2} (T_{k+1} s_k) s_k.$$

Rearranging and multiplying both sides by 3 for the first equation, and taking the inner product with s_k for the second equation, we have

$$3(f(x_k) - f(x_{k+1})) + 3 s_k^\top \nabla f(x_{k+1}) = \frac{3}{2} s_k^\top B_{k+1}^{\mathrm{C}} s_k - \frac{1}{2} s_k^\top (T_{k+1} s_k) s_k,$$
$$-s_k^\top y_k = -s_k^\top B_{k+1}^{\mathrm{C}} s_k + \frac{1}{2} s_k^\top (T_{k+1} s_k) s_k.$$

By summing up and using $y_k = \nabla f(x_{k+1}) - \nabla f(x_k)$, we eliminate the tensor term and obtain the scalar identity

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) + \frac{1}{2} s_k^\top y_k = \frac{1}{2} s_k^\top B_{k+1}^{\mathrm{C}} s_k.$$

Again, for some scalar $\sigma_k^{\mathrm{C}} \in \mathbb{R}$, we assume $\left(B_{k+1}^{\mathrm{C}} + \sigma_k^{\mathrm{C}} I\right) s_k = y_k$. Then, the previous equation yields

$$3(f(x_k) - f(x_{k+1})) + \frac{3}{2} s_k^\top (\nabla f(x_{k+1}) + \nabla f(x_k)) = -\frac{\sigma_k^{\mathrm{C}}}{2} \|s_k\|^2,$$

which means

$$\sigma_k^{\mathrm{C}} = -\frac{6(f(x_k) - f(x_{k+1})) + 3 s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2}.$$

Thus, the modified secant condition becomes

$$B_{k+1}^{\mathrm{C}} s_k = y_k + \frac{6(f(x_k) - f(x_{k+1})) + 3 s_k^\top (\nabla f(x_k) + \nabla f(x_{k+1}))}{\|s_k\|^2} s_k.$$

This is a cubic-augmented modified secant condition. Refer to Figure 16 for an illustration.

4.2 Other Curvature Related Methods

There are several other topics related to obtaining the curvature pairs s_k, y_k. Agg-BFGS [24] is an alternative approach to managing curvature information by aggregating data rather than discarding the oldest information and adding the newest. Multi-Secant [25] extends the secant condition framework by maintaining multiple pairs of step and gradient difference vectors. In standard form we have seen, it is defined as

$$s_i = x_{i+1} - x_i, \quad y_i = \nabla f(x_{i+1}) - \nabla f(x_i). \quad (i = k - m, \ldots, k)$$

In contrast, Multi-Secant centers all vectors around the latest iterate:

$$s_i = x_{k+1} - x_i, \quad y_i = \nabla f(x_{k+1}) - \nabla f(x_i). \quad (i = k - m, \ldots, k)$$

This approach can lead to better approximations in some cases.

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